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Abstract

This research investigates the potential of Robotic Process Automation (RPA) to improve the efficiency of the patient discharge process in Sri Lankan private hospitals. Patient discharge is a critical administrative task that often suffers from delays due to fragmented communication, and manual paper-based procedures that negatively affecting patient experience and hospital operational performance. Despite the growing global adoption of RPA in healthcare, its applicability within the Sri Lankan private healthcare context remains under-researched. The study uses a qualitative research approach, combining a systematic literature review, analysis of current discharge workflows, and case-based evaluation of RPA based practices from healthcare and related industries. Primary data is collected through questionnaires and interviews with hospital administrative staff, as well as patient feedback surveys, to identify inefficiencies, process bottlenecks, and readiness for automation. Based on these findings, the research proposes an RPA based discharge framework that identifies automation opportunities while clearly separating tasks requiring human judgment and clinical oversight. Rather than promoting a single automation platform, the study conducts a context based comparison of leading RPA technologies and supporting tools, examining their suitability under different organisational, technological, and financial conditions. This allows to develop a practical implementation guidance tailored to varying hospital environments. No software artefact is developed instead the study delivers an implementable strategic model for RPA adoption aligned with Sri Lanka's private healthcare environment. Ethical, legal, and data protection considerations including workforce implications, transparency, regulatory compliance, and patient data privacy are examined to ensure responsible and sustainable automation. The findings contribute both academically, by extending RPA research to a new regional context, and practically, by offering actionable insights for healthcare administrators considering automation to enhance discharge efficiency and patient experience.

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Glossary of Terms

- AI – Artificial Intelligence
- API – Application Program Interface
- GDPR – General Data Protection Regulation
- HIPAA – Health Insurance Portability and Accountability Act
- IA – Intelligent Automation
- IT – Information Technology
- ML – Machine Learning
- MPAR - Medication Prescription Administration Record
- NLP – Natural Language Processing
- OCR – Optical Character Recognition
- ROI – Return on Investment
- RPA – Robotic Process Automation
- PCO – Patient Care Officer
- UI – User Interface

1. Introduction

The private healthcare sector in Sri Lanka benefits many Sri Lankans throughout the country by offering speedier and more accessible medical care to everybody. Administrative activities are critical in hospital operations because they ensure that consumers receive efficient service. These responsibilities include patient registration, appointment scheduling, staff coordination, and, most significantly, patient discharge.

Among these administrative chores, one of the most essential is patient discharge, which has a direct impact on hospital bed availability, operational expenses, and patient satisfaction. Many Sri Lankan private hospitals execute this process manually, which involves a lot of paperwork, coordination across different divisions, and multiple clearances, resulting in delays and inefficiencies.

Robotic Process Automation (RPA), a rapidly developing technology noted for its capacity to automate rule based and repetitive manual activities, has demonstrated effectiveness in a variety of industries, including healthcare. These successful RPA implementations have proved increased efficiency, reduced human error, and improved service delivery. These success stories highlight the potential for RPA to be used in Sri Lanka's private healthcare industry, particularly to streamline and optimize the patient discharge process.

1.1. Background

Sri Lankan healthcare sector is divided as public and private sector. While public healthcare is free for all Sri Lankan citizens, it has various challenges such as overcrowding, long waiting times and shortage of resources and staff. As a result, majority of the population choose the private sector for their health needs. The private healthcare sector handles over 60% of the consultations when compared to the public sector highlighting that people select private sector for a faster and efficient service. (Karunapema & Abeykoon, 2016)

As of 2025 there are 72 registered Private Hospitals, Nursing Homes, and Maternity Homes across Sri Lanka, according to the Private Health Services Regulatory Council (Private Health Services Regulatory Council, 2025). Many of these hospitals have a number of branches island wide which serve hundreds of patients on a daily basis. These patients are admitted for various reasons such as surgeries, emergencies, long-term care or consultancy. During these visits or stays some patients often interact with several consultants and gain numerous services, which must be considered at the time of discharge.

The discharge process in many of these hospitals requires the discharge unit to coordinate with multiple departments to collect documents, service rates, and discharge notes. This coordination is done manually most of the time, which causes delays which leads to delayed hospital stays. These inefficiencies affect bed availability, increased operational costs, and result in patient dissatisfaction.

RPA, also known as Robotic Process Automation, is a technology that uses software robots or "bots" to automate repetitive, rule based tasks performed by humans, such as data entry, form filling, file transfers, and other routine based processes. RPA mirrors human actions by interacting with systems through user interfaces (UI) or APIs. It aims to improve efficiency, reduce errors, and frees up employees to focus on value adding tasks. RPA can be improved with Intelligent Automation (IA), which includes Artificial Intelligence (AI), machine learning, and natural language processing to handle more complex tasks. While RPA follows predefined rules, AI allows the software bots in decision making and learning from data. Industries like banking, healthcare, retail, and insurance widely use RPA to streamline their operations and cut costs. (IBM, 2022)

1.1.1. Current Approaches & Limitations

Currently, the discharge process in most Sri Lankan private hospitals is done manually. Once the doctor gives the clear for the patient to go home, the discharge unit requests and gather necessary information from all relevant departments such as billing, pharmacy, and consultants. This process includes finalizing the total charges, collecting discharge notes of the patient, and ensuring all treatments and procedures are recorded in the hospital systems. This back and forth communication often involve many paper based forms which need to be entered into the system, phone calls to relevant departments, or waiting on staff availability, leading to a time consuming process and a lack of visibility of the discharge status to the patients.

Such ineffectiveness not only delay patient discharge but also impact on bed or room occupancy issues. Additionally, due to the absence of a centralized or automated system, most of the data entry in this process is done manually, which is both time consuming and leads to too many errors.

1.1.2. Significance of the Project

RPA has proven to successfully streamline repetitive tasks across the world, this includes rule based administrative tasks in healthcare. Applications of RPA in healthcare include automating the billing process, consolidating patient records to one system, and generating discharge summaries reducing human involvement while increasing speed and accuracy.

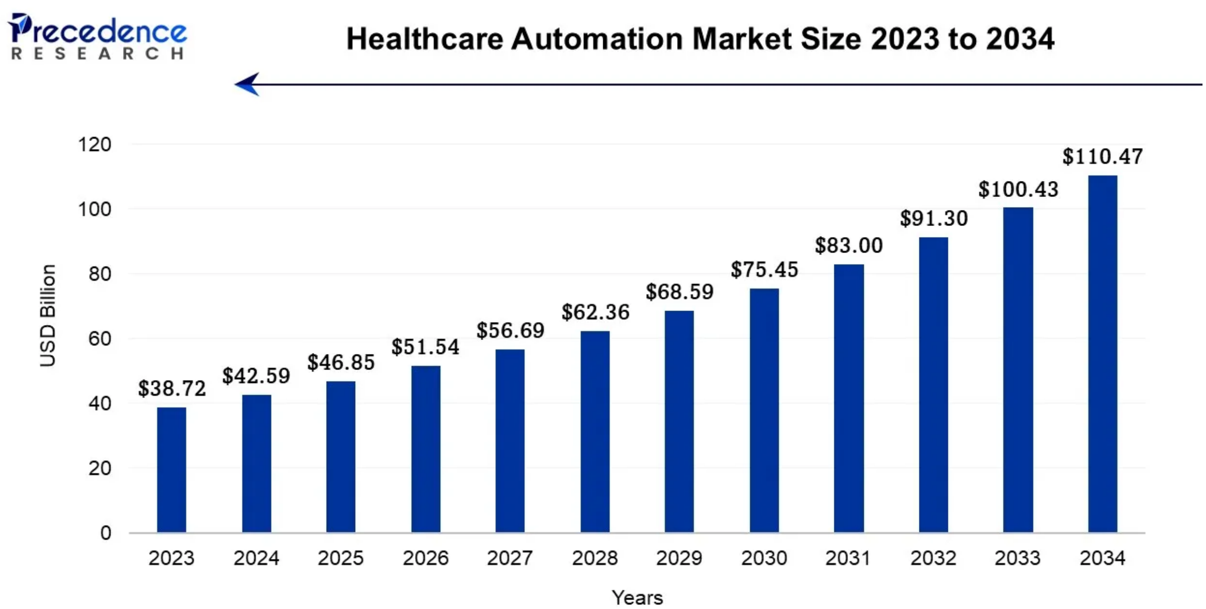

Figure 1: Healthcare Automation Market Size 2023 to 2034 (Precedence Research, 2024)

It is clear that the market for healthcare automation is growing significantly in the years to come, and it is expected to reach around USD 110.47 billion by 2034. (Precedence Research, 2024). Therefore, to meet this demand, Sri Lankan healthcare sector also needs to implement such automation to provide patients with efficient and faster services.

In many counties these automation is being placed in hospital and was able to increase efficiency in a considerable rate. Given the increasing patient demand in Sri Lanka's private hospitals and the current ineffectiveness in discharge process, the adoption of RPA could improve hospital operations.

1.2. Motivation

Currently in Sri Lanka there is a demand for quality healthcare services by the private sector this demand highlights the need for private hospitals to streamline their administrative processes. Among these said processes, the patient discharge process plays a vital role, as it directly affects the bed availability, hospital efficiency, and the overall the patient experience as it is the final experience the patients will receive during their hospital stay. However, currently because of the manual nature of this process in most private hospitals it continues to create bottlenecks, delays, and dissatisfaction.

Globally, many healthcare institutions are rapidly implementing RPA to automate their tasks, to improve data accuracy, and speed up the processes. These success stories, especially in countries with advanced healthcare systems, show that RPA can significantly reduce administrative burdens and improve hospital performance.

Regardless of these proven successes, Sri Lanka's private healthcare sector has not adopted this technology, specifically in the patient discharge process. Therefore, this research is motivated by the opportunity to explore the feasibility and impact of introducing RPA in optimizing the patient discharge process in Sri Lanka. Additionally, motivation is increased due to the growing trend in healthcare automation worldwide by 2034.

1.3. Research Problem

Sri Lanka's private hospitals currently rely deeply on manual administrative tasks for discharging patients. This manual process involves communicating between multiple departments, paper based documentation, and repetitive data entry, which leads to delays, errors, and inefficiencies. These inefficiencies negatively impact on hospital operations from increasing wait times and reducing patient satisfaction.

As global case studies display the potential of RPA to improve hospital administration, there is a significant gap in localized research or implementation efforts within the Sri Lankan private healthcare sector. Additionally, there is no clearly defined structure or solution tailored to address the inefficiencies in the Sri Lankan discharge process through RPA.

Therefore, the research is planning to investigate whether and how RPA can be implemented in the Sri Lankan private healthcare sector to improve the patient discharge process. Which will include in identifying the inefficiencies in current process, assessing the technological feasibility, and proposing a practical RPA based solution.

1.4. Aims and Objectives

1.4.1. Aim

To enhance the efficiency and effectiveness of the patient discharge process in Sri Lanka's private healthcare sector through the implementation of RPA.

1.4.2. Objectives

1. Inspect the current patient discharge process of private hospitals in Sri Lanka and identify inefficiencies.

    - Conduct an analysis of patient discharge workflows in private hospitals over a three week period.

    - Identify inefficiencies such as delays, data entry errors, and inter-department coordination issues within one additional week, based on stakeholder interviews and process documentation.

2. Explore the potential of RPA in optimizing the patient discharge process.

    - Review a minimum of twenty academic articles and five industry reports on RPA applications in healthcare over a three week period.

    - Analyse three to five global case studies of RPA implementation in hospitals within 1 week after the literature review.

3. Propose an RPA based solution to automate manual repetitive tasks in the discharge process.

    - Design a conceptual model for RPA integration using process diagrams and flowcharts within 2 weeks.

    - Identify which tasks can be automated vs. those needing human intervention in one additional week based on workflow analysis.

4. Assess the feasibility of RPA adoption in Sri Lanka's private healthcare sector.

    - Identify potential technological, financial, and regulatory challenges over a two week period.

    - Evaluate the expected impact of RPA on hospital operations and patient experience in three weeks using patient feedback.

5. Develop recommendations for private hospitals considering RPA adoption.

    - Prepare a recommendation report including guidelines for implementation, anticipated benefits, and best practices within 3 weeks.

    - Highlight best practices for successful RPA implementation while maintaining compliance with healthcare regulations.

1.5. Research Contributions

This dissertation makes several contributions to both Information Systems research and professional practice. First, it provides one of the first research of Robotic Process Automation applicability within Sri Lankan private healthcare, addressing a clear geographical and contextual gap in the existing literature. Second, the study develops a context specific, RPA based patient discharge framework that combines process re-engineering, organisational readiness, and governance considerations rather than treating automation as a purely technical solution. Third, the research introduces a decision oriented perspective on RPA platform suitability, showing how organisational context, digital maturity, and system complexity should guide tool selection rather than vendor preference. Finally, the study contributes to Information Systems knowledge by establishing the importance of socio-technical alignment in healthcare digital transformation initiatives within developing countries where process maturity and regulatory constraints play an important role.

2. Literature Review

2.1. Introduction

The efficiency of the patient discharge process is a key factor in a hospital that directly affects the patients' return rate and quality of service. In many hospitals, the discharge process starts from the final medical clearance for the patient to leave until the final bill payment. This process involves coordination among doctors, nurses, administrative staff, and insurance providers. Delays in this process are recognized as major bottlenecks that increase the operational costs, prolonged bed consumption (Simbo, Inc, 2025), and patient dissatisfaction (Kumar, 2022). RPA has risen in the field of Information Technology (IT) in recent years as a promising solution that streamline such administrative workflows. RPA involves deploying software "bots" to impersonate human interactions with information systems, automating repetitive, rule based tasks in healthcare, and is already being implemented in hospitals for functions like scheduling, billing, claims processing and electronic record management around the world.

However, despite reported efficiency gains, most existing studies focus on digitally mature healthcare systems, limiting their applicability to heavily paper based hospital environments that are common in developing countries. The literature contains few studies on RPA applications in South Asian hospitals, particularly in Sri Lanka. Despite having a relatively well developed healthcare system across both public and private sectors, the implementation of RPA in Sri Lanka remains largely unexplored in both academic research and industry practice.

This section reviews the patient discharge process in private hospitals, the concept and evolution of RPA, its applications in healthcare, and the challenges associated with its implementation. The review evaluates global evidence while highlighting contextual factors relevant to Sri Lanka, ultimately identifying research gaps that justify the need for the present study.

2.2. The Patient Discharge Process in Private Hospitals

The patient discharge process typically begins when the doctor who is treating the patient determines that he or she is ready to leave. It is then followed by a sequence of tasks, including finalising the discharge summaries, arranging follow-up appointments, dispensing discharge medications, completing billing and insurance procedures, and providing the patient or caregiver with guidance on homecare. These steps often involve coordination among multiple departments. For instance, doctors must write medical summaries, nurses compile clinical records of the patient, pharmacists prepare discharge medications, and administrative staff process the billing. (Paula R. Patel, 2025) (Emes M, 2018)

This discharge process is a standard procedure followed around the world, which is prone to inefficiencies. Studies have reported that the traditional discharge procedure is frequently delayed due to administrative blockers such as paperwork errors, incomplete documentation, and multiple approvals across departments. An assessment on a discharge practice in multiple hospitals were observed and it was found out that discharges typically take between 4 to 5 hours from the time the doctor clears a patient. (Kumar, 2022) For example, in an Indian hospital study covering 25 to 30 daily discharges showed that cash paying patients experienced an average delay of six hours, while those with insurance waited over seven hours, due to billing, pharmacy, and clearance of files from various departments. (Saurabh Sharma, 2024) These delays lead to prolonged bed occupancy, which in return limits bed availability for new admissions and reduces overall hospital throughput. (Simbo, Inc, 2025)

In the Sri Lankan private healthcare sector which comprises multiple hospitals across the country has more than 135,000 annual admissions means they too face the above mentioned discharge inefficiencies due paper based systems these hospitals follow. (Dissanayaka, 2023) Which includes forms such as admission records, lab results, medication charts, and discharge summaries which are typically processed manually, creating notable administrative burdens on hospital staff.

Though local data on average discharge times in Sri Lanka is limited, international case studies from similar healthcare contexts suggest that significant efficiency improvements are achievable through process re-engineering. One case study from a sixty bed Nigerian hospital, where manual discharge delays were dominant, documented a 44% reduction in discharge time following a process streamlining intervention. (Umelo FU, 2025) Taken together, these findings suggest that discharge delays are predominantly administrative rather than clinical, reinforcing the suitability of automation based interventions. However, successful implementation requires careful process optimisation prior to automation.

2.3. Robotic Process Automation Concepts and Evolution

Robotic Process Automation refers to software technologies that use software bots or scripts to mimic and automate human interactions with digital systems. These bots can log into applications, extract or enter data, move files, and complete forms just as an office worker would (IBM, 2022). RPA combines user interface actions and APIs to perform tasks across multiple systems and execute large volume of processes with high accuracy (Aguirre & Rodriguez , 2017). In other words, while a staff might spend minutes updating systems manually, an RPA bot can complete the same work instantly without errors. (Wewerka & Reichert, 2020)

Unlike artificial intelligence, RPA is primarily process driven, these bots follow precise, pre-defined set of rules which limited its applications. Due to this modern RPA platforms started integrating AI components into RPA, creating intelligent automation. This new enhancement can manage semi-structured data unlike traditional RPA (Ribeiro, et al., 2021) (Deo & Anjankar, 2023). Leading vendors (such as UiPath, Automation Anywhere, Power Automate, etc.) now bundle RPA with machine learning models, optical character recognition, and natural language processing so bots can handle more complicated tasks. This evolution allows RPA to grow from simple task automation tools into broader digital workforce platforms. Still, RPA’s core remains as a rule-based automation and therefore, cannot replace human judgment or creative problem solving. (UiPath, 2023) (IBM, 2022)

2.4. Applications of RPA in the Healthcare Sector

The healthcare sector has seen a growing adoption of RPA across both administrative and clinical support areas. On the administrative side, RPA has been widely applied to patient registration, appointment scheduling, billing, and claims processing (Chaturvedi & Sharma, 2023). For example, UiPath reports that RPA bots are used to automate patient registration, insurance claims submission, and audit management, leading to improvements in processing speed and data accuracy (UiPath, 2018). In appointment scheduling, bots can review appointment requests, apply predefined rules such as doctor availability, insurance coverage, and service location, and generate optimised schedules for call centres, thereby reducing backlogs and improving patient satisfaction.

Finance management is another area where RPA has proven benefits. Automation Anywhere documents that St. John of God Health Care in Australia implemented RPA for billing and receipting operations, allowing the processing of nearly one billion AUD in transactions annually and saving approximately 25,000 staff hours (Automation Anywhere, 2020). Similarly, RPA bots have been used to automate accounts payable and receivable processes, notify patients of outstanding balances, and accelerate payment collection when compared to manual methods.

Relevance to this study, RPA has also been applied directly to patient discharge workflows. Automation bots can verify discharge documentation, validate medication lists, confirm follow-up orders, and issue automated reminders for post discharge appointments or prescription refills. UiPath mentions that such automation helps ensure the accuracy and completeness of discharge instructions and alerting caregivers only when further intervention may be required (UiPath, 2018). In practice, organisations like Cleveland Clinic have deployed RPA for discharge management through an automated bot that processes discharge reviews after hours within the electronic health record system. As a result, nursing staff no longer face discharge backlogs at the start of the day, with approximately 45–52% of monthly discharge reviews automated through RPA. (Cleveland Clinic, 2024)

Research examining the application of RPA in medical record management found that routine documentation tasks, including discharge summaries, billing, and insurance verification, were completed up to 70% faster following automation, while error rates decreased from approximately 5% to below 1%, resulting improved data integrity and operational reliability (James Andrew, 2024). In addition, a recent West Asian case study confirmed that integrating RPA with Lean Six Sigma methodologies reduced hospital process lead times by approximately 380 minutes and improved Process Cycle Efficiency from 69% to 95.5%, showing the potential of RPA to improve both speed and process quality when combined with structured improvement frameworks (Huang, et al., 2024).

Beyond discharge activities, RPA has been applied to post-care coordination and follow-up processes. Automation Anywhere describes scenarios in which bots automatically schedule home healthcare visits, arrange follow-up services based on discharge orders, and monitor patient compliance with outpatient appointments and diagnostic tests. Automated alerts and reminders allow more consistent continuity of care, demonstrating how RPA can support end-to-end process from inpatient discharge to outpatient follow-up (Automation Anywhere, 2024).

Despite these reported benefits, the majority of existing evidence comes from technologically mature healthcare environments which has established electronic health record systems and standardised workflows (Automation Anywhere, 2020) (Cleveland Clinic, 2024). These studies focus on post implementation performance metrics while offering limited insight into the organisational readiness, process standardisation efforts, and governance mechanisms required prior to automation. Additionally, vendor led case studies often highlight operational efficiency gains while underselling challenges related to system instability, long term maintenance requirements, and workforce adaptation. Among the reviewed literature, the implementation at National Taiwan University Hospital provides one of the closest backgrounds to Sri Lanka, showing good time savings after RPA adoption (Huang, et al., 2024). However, even this case operates within a more digitally mature environment than most Sri Lankan private hospitals.

Therefore, while the literature clearly confirms RPA's technical potential to improve discharge related and administrative healthcare processes, it offers limited data on how such solutions can be adapted to environments with partial digitisation, mixed system infrastructure, and varying levels of organisational readiness, such as those found in Sri Lanka. This limitation emphasizes the need for context aware automation frameworks rather than direct replication of solutions from technologically advanced healthcare systems.

2.5. Challenges and Limitations of RPA Implementation in Healthcare

Despite its promising nature, RPA implementation in healthcare faces several challenges. Data privacy and security are one of the main such concerns, access to patient records must comply with the Health Insurance Portability and Accountability Act (HIPAA) and other regulations. RPA in healthcare requires a good technical knowledge and must address data privacy concerns, high implementation costs, and complex regulatory requirements. These factors can result to hospitals not adopting RPA, especially in environments with sensitive health information. (Nimkar, et al., 2024)

Hospitals also worry about integrating RPA tools into legacy IT systems. In many healthcare settings, information is stored in older applications which do not have modern APIs and UIs. RPA bots often rely on user interfaces, so any change in the software UI can disrupt the functionality of the RPA bot and force time consuming bot reconfiguration For example, if a system upgrade moves data fields, the bot may fail until the bot is reprogrammed. This fragility and maintenance overhead are significant barriers. Another limitation is that RPA handles only structured, rule-based tasks. RPA encounters limitations with tasks requiring human empathy or judgment. In healthcare, many processes involve unstructured data such as free text notes, variable patient conditions or tasks that need clinical decision making. Bots cannot replace those. (Kraus, et al., 2024)

Scaling RPA is also challenging, a solution that works for a small process may not easily generalize to complex or large scale workflows, especially in fast growing hospital operations. Finally, human factors matters some staff may fear that RPA threatens jobs. (Nimkar, et al., 2024) (Redwood

Software, 2024) In summary, RPA adoption requires careful planning addressing data governance, ensuring stable IT environments, and involving clinical stakeholders to overcome these hurdles.

Existing literature highlights data privacy, legacy system compatibility, and workforce resistance as primary barriers to RPA adoption in healthcare (Redwood Software, 2024) (Nimkar, et al., 2024). However, many studies treat these challenges as secondary implementation issues rather than fundamental design constraints. In situations where paper based documentation, fragmented systems, and process variations are dominant, automation without prior process re-engineering may increase inefficiencies rather than resolve them. This limitation is particularly relevant for developing healthcare systems, where digital maturity varies across institutions. Therefore, the literature suggests that successful RPA adoption is less dependent on the advance nature of automation tools and more dependent on process standardisation, governance frameworks, and human oversight systems. This understanding directly influenced the design of the proposed solution in this study.

2.6. Research Gaps and Justification for the Study

The most significant gap identified in the literature is the near absence of local studies on RPA adoption within Sri Lankan healthcare. Existing research primarily focuses on developed countries with mature digital infrastructures, offering limited insight into automation feasibility in paper-based hospital environments. Beyond the geographical gap, there is also a notable process level gap, as existing studies rarely examine end to end patient discharge workflows, focusing instead on isolated administrative tasks such as billing or claims processing. Furthermore, limited research addresses the preconditions required for successful RPA adoption, including process re-engineering and digital data standardisation.

Accordingly, this study is justified by both academic and practical needs. By examining RPA feasibility within Sri Lankan private hospitals, this research extends global automation knowledge to an underexplored context and provides actionable insights for healthcare administrators and policymakers.

2.7. Summary

In summary, existing literature demonstrates that RPA can effectively automate repetitive healthcare processes, including elements of patient discharge management. International evidence confirms improvements in efficiency, accuracy, and staff workload reduction. However, challenges related to data privacy, legacy system integration, and organisational readiness persist.

Importantly, the Sri Lankan private healthcare sector has yet to adopt RPA, representing both a challenge and a significant opportunity. The reviewed literature supports the need for context sensitive research that considers process re-engineering and phased automation adoption. These insights shape the proposed RPA solution and support the discussion on whether Sri Lankan hospitals are ready to implement it.

3. Methodology

3.1. Research Approach and Design

This study adopts a qualitative research approach to investigate the feasibility and suitability of applying RPA to the patient discharge process in Sri Lankan private hospitals. A qualitative methodology is considered appropriate due to the process focused, organizational, and socio-technical nature of the research problem, where understanding the workflow, stakeholder interactions, and contextual constraints is more critical than statistical measurement.

The research design combines a systematic literature review, primary qualitative data collection (patient surveys and staff interviews/questionnaires), and comparative analysis to examine current discharge practices and identify opportunities for automation. A case based evaluation approach is used to contextualize findings within the operational constraints of Sri Lankan private healthcare institutions, rather than targeting for generalisation.

Quantitative or experimental methods were not adopted, as the study does not seek to measure performance outcomes or test a deployed system, but rather to explore possibility, readiness, and design considerations for RPA adoption. The proposed solution remains conceptual, focusing on process redesign and automation potential than technical implementation.

Methodological accuracy is supported through data triangulation, drawing insights from literature, patient perspectives, and healthcare staff experiences to improve the reliability of the findings. This approach guarantees alignment between the research objectives, data collection methods, and the nature of the proposed RPA enabled discharge workflow.

3.2. Technologies and Resources

Although the project involves no practical development, a detailed study of leading RPA technologies and tools will be conducted to give out a logical solution that can be implemented in hospitals. Therefore, multiple leading RPA technologies will be explored to identify their key features, use cases, integration potential, and adaptability. This will support the design of a feasible, automation framework to optimize the discharge process.

3.2.1. Technologies to be Explored

RPA Platforms

Several widely adopted RPA platforms will be examined, focusing on their capabilities in healthcare contexts:

- UiPath: One of the RPA tools known for its enterprise level adoption and user friendly visual workflow interface. This tool offers broad automation solutions including document processing and integration with AI.

- Automation Anywhere: This solution is a cloud based RPA tool with strong built in AI capabilities and a focus on intelligent automation in industries like healthcare.

- Blue Prism: Known for secure, scalable RPA deployment across large enterprises. Blue Prism mostly focuses on compliance, central governance, and auditability.

- Microsoft Power Automate: This is a product of Microsoft that helps integration with Microsoft tools (Excel, SharePoint, Outlook), making it highly adaptable in administrative healthcare workflows where the IT infrastructure is based on Microsoft products.

Supporting Technologies

To enhance the effectiveness and intelligence of RPA, the following complementary technologies will also be reviewed:

- Artificial Intelligence (AI) & Machine Learning (ML): These technologies can be useful in predictive analytics and decision support systems within discharge processes.

- Natural Language Processing (NLP): NLP can help in automation of tasks involving communication, such as interpreting discharge instructions or processing email-based approvals.

- Optical Character Recognition (OCR): Support is converting handwritten or printed documents such as discharge summaries, medical prescriptions into machine-readable formats.

3.3. Data Sources

The research will be based on both primary and secondary data sources, ensuring a good understanding into the patient discharge process and automation feasibility within Sri Lankan hospitals.

Primary Data

- Semi Structured Questionnaires and Interview: Interviews were conducted using a semi structured questionnaire from administrative staff and healthcare professionals in a selected Sri Lankan private hospital which will help to gain direct insights into discharge process challenges, RPA feasibility, and hospital specific constraints.

- Patient Feedback Surveys: A structured form was distributed to recent patients from private hospitals to capture their experiences on discharge delays, communication issues, and documentation inefficiencies.

The data collected through these two methods is used for analysis presented in Chapter 4, which will look into key patterns, experiences, and opportunities for RPA adoption within Sri Lankan private hospitals.

Secondary Data

- Academic Literature: Peer reviewed studies and systematic reviews related to RPA in healthcare, with particular focus on administrative process automation.

- Industry Reports: Global and regional reports from Deloitte, EY, McKinsey, Gartner, Forrester, etc. will provide standards, ROI metrics, and trend analysis on healthcare automation.

- Case Studies: Case studies from industries like banking, logistics, and insurance sectors with more RPA implementations will be analysed to identify best practices, challenges and other factors that might support in RPA implementation in healthcare.

3.3.1. Sampling and Participants

Based on the two primary data sources the below sample sizes were selected,

1. Staff Questionnaire: 20 hospital staff members from a leading private hospital in Colombo were selected, comprising three participants with varying seniority levels (senior, mid, and junior) from the six departments directly involved in the discharge process. Ward Nursing, Ward Billing, Ward Doctor, Ward Patient Care Officer (PCO), Pharmacy, and Laboratory along with two participants from the Admissions department to provide insights into the initial phase of the discharge workflow.

2. Patient Feedback Surveys: 100 people who recently discharged from a private hospital in Sri Lanka or was involved in a discharge process of a family member.

3.3.2. Participant Selection Process

Prior to interviews with hospital staff, an initial discussion was held with the Hospital Operations Manager of a leading private hospital in Sri Lanka to identify the departments that are involved in the patient discharge process. Based on this, six departments were identified as directly contributing to discharge process such as, Ward Nursing, Ward Billing, Ward Doctor, Ward Patient Care Officer (PCO), Pharmacy, and Laboratory. From each of these departments, approximately three staff members were interviewed, for a balanced mix of operational and administrative perspectives. In addition, two participants from the hospital admission department were included to gain insights into the starting point of the discharge workflow, for a complete process understanding.

For the patient survey, the questionnaire was distributed among individuals who had been discharged from private hospitals located in Sri Lanka or was involved in a discharge process of a close family member. This ensured that all patient participants had experience with the discharge process within private hospital settings, for a better context specific feedback.

3.3.3. Sample Size Justification

In qualitative research, the goal is to obtain detailed insights rather than statistical generalization. Therefore, a total sample of 120 participants including of 100 patients and 20 staff members was considered sufficient to reach data saturation. The larger patient group allowed to identify recurring patterns in discharge experiences, while the smaller but targeted staff group provided an in depth operational view from different departments involved in the discharge process. These combined responses offered both detailed and comprehensive data, allowing to study the difference between patient and staff viewpoints. Given that qualitative studies commonly achieve thematic saturation with smaller samples, this study's sample size is believed to be sufficient to ensure the reliability and credibility of the findings within the Sri Lankan private healthcare context.

4. Data Analysis & Findings

This chapter presents and analyses the findings obtained from the Patient Discharge Experience Survey and the Staff Questionnaire. The analysis explores how participant responses reveal existing inefficiencies, coordination gaps, and automation opportunities within the patient discharge process of Sri Lankan private hospitals. A qualitative analytical methodology is used, combining descriptive interpretation of survey data with thematic insights derived from participant responses. Graphical representations are used to illustrate dominant patterns, while informational comments are used to contextualise the findings.

4.1. Patient Survey Findings

This section focuses on insights from the Patient Discharge Experience Survey conducted among patients and family members who had recently experienced discharge from private hospitals in Sri Lanka. The survey aimed to understand discharge duration, recognized sources of delay, communication effectiveness, and patient attitudes towards digitalisation and automation. The findings provide an external, user focused view of discharge inefficiencies and help identify areas where automation could enhance transparency, efficiency, and patient satisfaction.

4.1.1. Background of Respondents

Although the survey initially targeted 100 participants, a total of 111 responses were received from patients or family members involved in a recent discharge at private hospitals in Sri Lanka. This exceeded the planned sample size and improved the credibility of the findings.

The demographic profile of respondents showed a diverse group of private hospital users. The majority of participants were discharged from hospitals located within the Colombo District, indicating the availability of large private healthcare providers in this region. Patients aged between 51 and 70 years was the largest age group, suggesting that older adults are among the most frequent users of private inpatient healthcare services. In terms of admission type, medical and surgical cases were the majority, followed by diagnostic or short-term observation admissions. Importantly, over 54% of respondents reported discharge experiences within the previous year, ensuring that the findings reflect current operational practices.

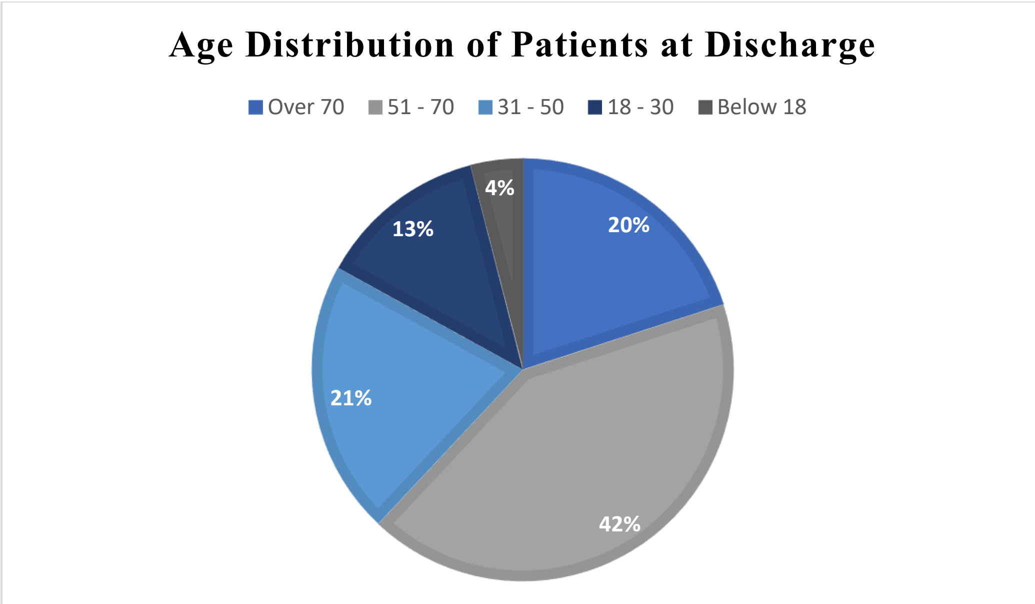

Figure 2: Age Distribution of Patients at Discharge

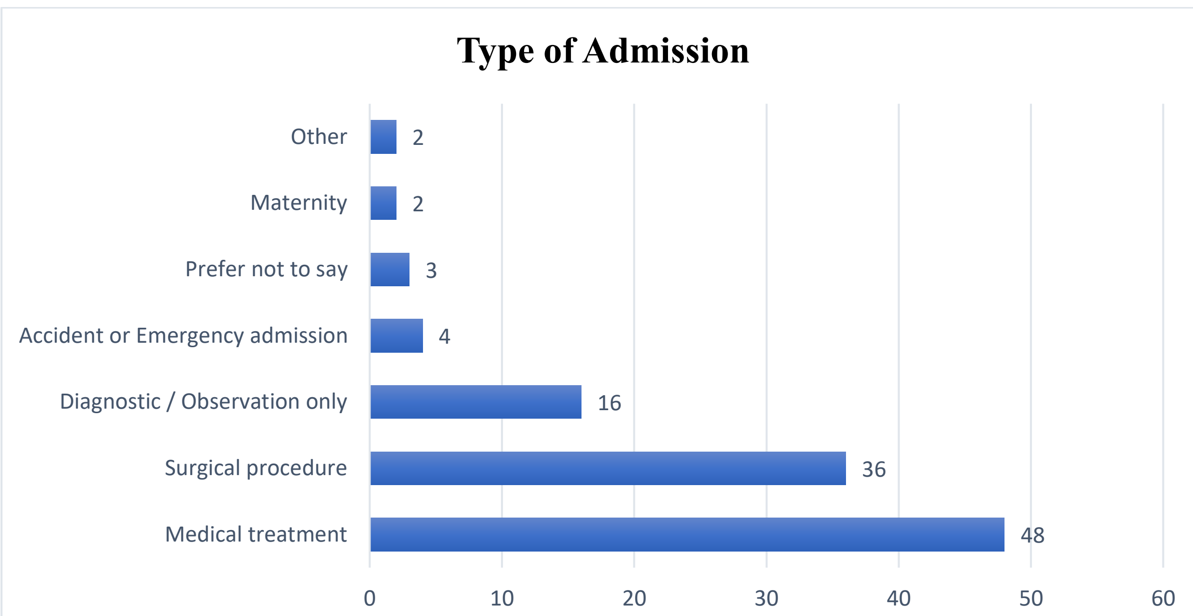

Figure 3: Type of Admission

Overall, the responders of the survey are from a realistic group of private hospital patients in Sri Lanka. The majority of medical and surgical admissions reveals discharge scenarios that are typically administration intensive rather than clinically complex. Therefore, highlights strong potential for automation, as many discharge related activities in such cases involve repetitive, rule based administrative tasks rather than medical decisions.

4.1.2. Discharge Experience and Delays

Participants were asked to report the waiting time experienced after receiving medical clearance from the attending physician. The findings indicate that discharge delays are common, with the majority of respondents reporting waiting times exceeding three hours. A considerable amount also reported delays of one to two hours, after being medically cleared.

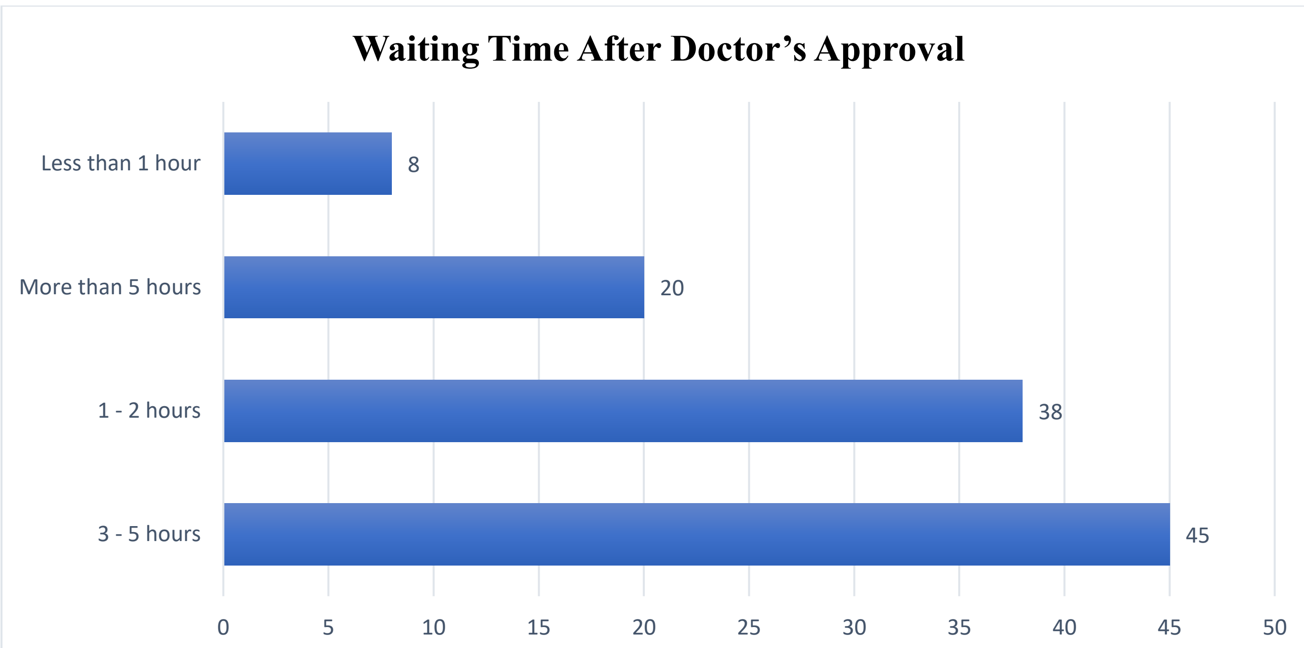

Figure 4: Waiting Time After Doctor’s Approval

Respondents were also asked to identify the primary causes of delay during discharge. The most frequently mentioned factors were billing procedures, discharge documentation receival, medication collection from pharmacy, and insurance related processing.

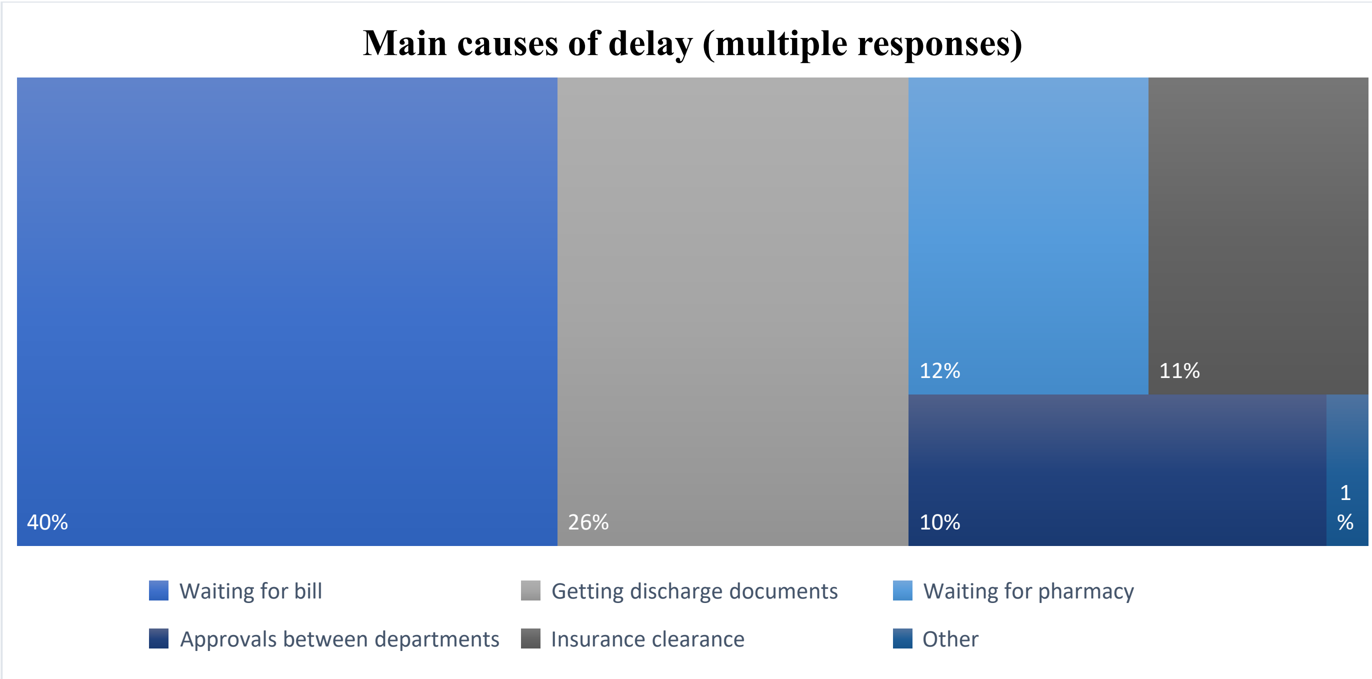

Figure 5: Main Causes of Delay During Discharge

These findings clearly indicate that discharge delays are largely due administrative and coordination related activities rather than clinical readiness. Tasks such as billing finalisation, document preparation, and pharmacy processing are largely manual and involve multiple departments, resulting in longer waiting times. This indicates fragmented workflows and limited process integration. The results suggest that administrative coordination represents the core inefficiency within the Sri Lankan private hospital discharge process and therefore presents a high opportunity for automation through RPA.

4.1.3. Communication and Process Organization

The survey also examined patient experiences of communication clarity and process organisation during discharge. While 74% of respondents said that discharge instructions and medication explanations were clearly communicated, nearly 21% reported that communication was only “somewhat” clear. Similarly, when asked whether the discharge process felt smooth and well organised, only 25% of respondents responded positively, with the remainder expressing neutral or negative perceptions.

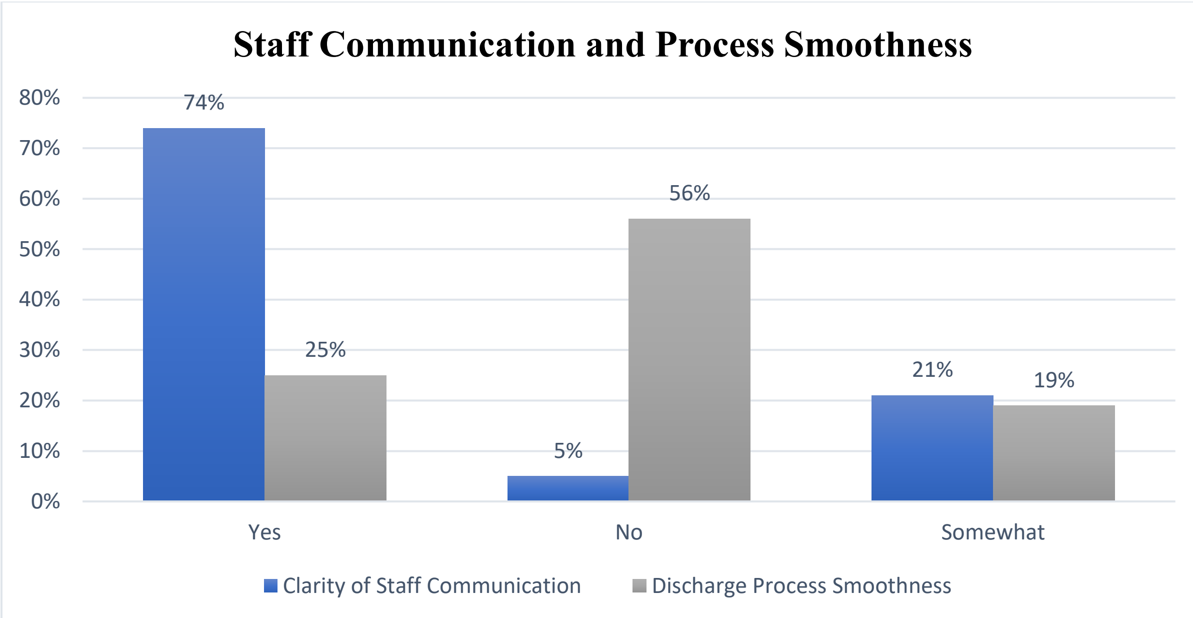

Figure 6: Clarity of Staff Communication During Discharge and Process Smoothness

These findings suggest that although hospital staff are generally professional and supportive, communication gaps still exist during the discharge process. Many patients reported uncertainty regarding waiting times, process progress, and next steps. This lack of visibility implies a process design limitation rather than a staff performance issue. The results highlight an opportunity for digital status notifications, automated updates, and workflow tracking systems to improve transparency and reduce patient anxiety during discharge.

4.1.4. Perception of Digitalisation and Automation

Patient attitudes towards digitalisation and automation were strongly positive. More than 75% of respondents believed that automation or digital solutions would improve discharge efficiency, while 82% indicated that they would actively support hospitals adopting such technologies.

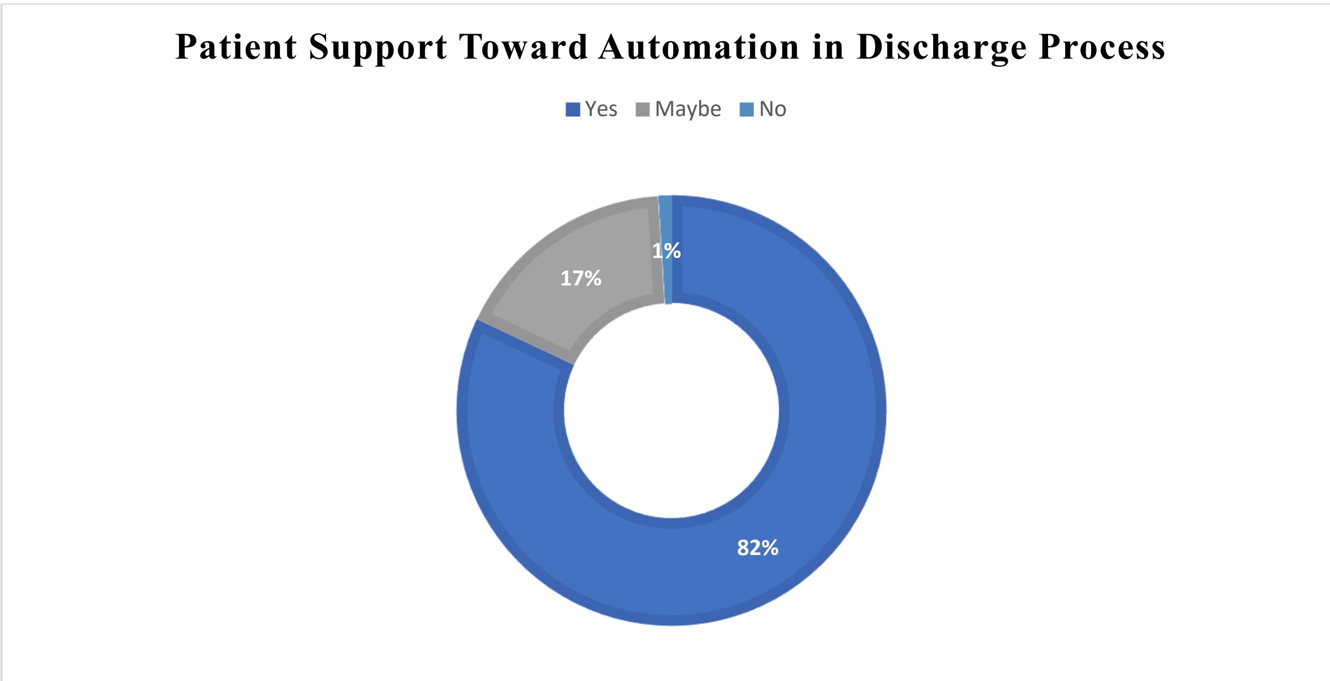

Figure 7: Patient Perception Toward Automation in Discharge Process

Qualitative comments further revealed that patients believe with automation they will achieve improved transparency, reduced waiting times, and greater convenience. Suggestions given frequently included digital billing, electronic documentation, and automated notifications. These responses show a high level of social acceptance and readiness for automation, reinforcing the feasibility of introducing RPA driven improvements within private hospital discharge workflows.

4.1.5. Summary of Patient Insights

The patient survey highlights dissatisfaction with discharge delays caused by billing, pharmacy, and insurance procedures. While patients value staff professionalism in private hospitals, they experience administrative blocks and lack of process visibility during discharge. The majority of the respondents express strong support for automation, believing it will improve speed, transparency, and service quality. These findings confirm the appropriateness of RPA as a solution for enhancing patient experience and operational efficiency within the discharge process.

4.2. Staff Interview and Questionnaire Findings

This section discusses findings from staff interviews and questionnaires conducted across six departments involved in the patient discharge process: Ward Nursing, Ward Doctors, Ward Billing, PCO, Pharmacy, and Laboratory, with additional insights from the Admission department. The objective of this analysis is to examine discharge related work distribution, operational bottlenecks, and staff perceptions toward automation.

Adopting a qualitative perspective, the analysis focuses not only on what delays occur, but why they occur by examining the interaction between people, processes, and supporting technologies within the discharge workflow.

4.2.1. Departmental Roles and Experience

The majority of participants were from departments directly engaged in discharge related activities on a daily basis, ensuring that responses reflected experiences rather than observations. Most respondents had more than three years of experience within the hospital environment, indicating strong familiarity with internal workflows and administrative procedures.

To capture variation in responsibility, interviews were conducted with staff across senior, mid-level, and junior roles within each department. This approach revealed that experience level plays a significant role in determining involvement in discharge activities, particularly within clinical units.

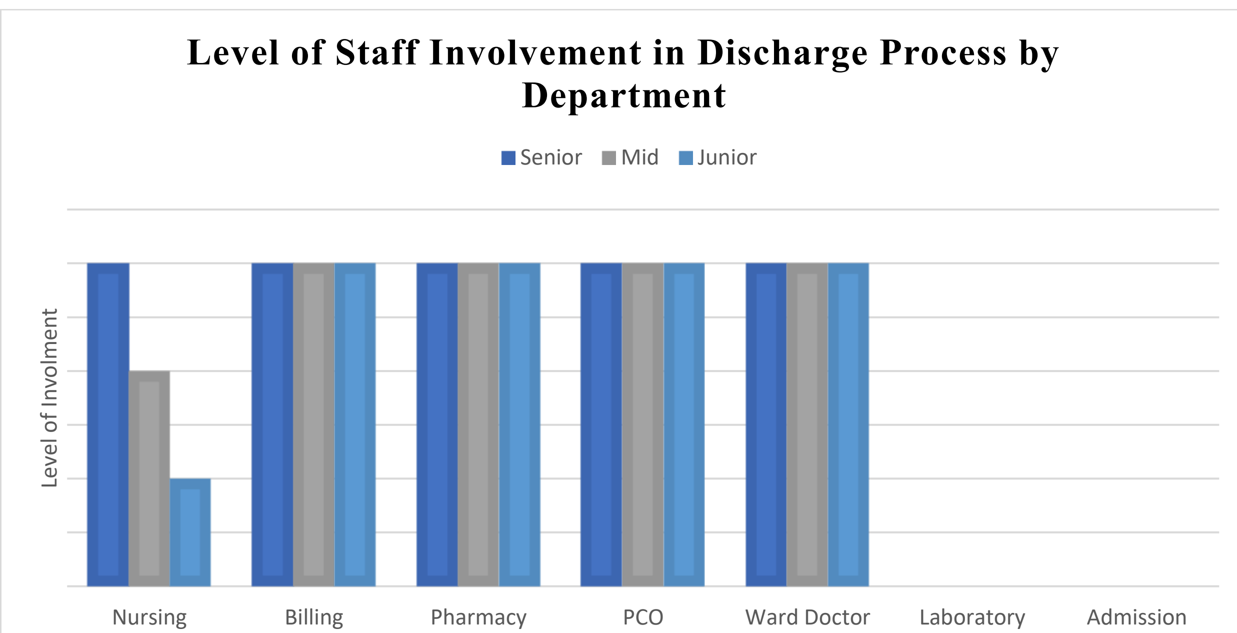

Figure 8: Level of Staff Involvement in Discharge Process by Department

As illustrated in Figure 8, Ward Nursing shows a highly experience dependent structure. The nurse-in-charge performs the majority of discharge related tasks, including documentation completion, charge validation, and coordination with billing. Mid-level nurses provide occasional support during peak workloads, while junior nurses are rarely involved. This differs with departments such as Pharmacy, Billing, and PCO, where discharge responsibilities are evenly distributed across staff regardless of seniority.

This characteristic indicates that some discharge activities are procedural and standardised, while others rely heavily on individual experience and knowledge. From an Information Systems perspective, experience dependent processes represent higher operational risk, as delays and inconsistencies arise when key individuals are unavailable. These findings highlight strong potential for automation in administrative discharge tasks that are rules based and repeatable, while highlighting the need for human oversight in clinically sensitive activities.

4.2.2. Current Discharge Workflow

Staff findings confirm that the discharge process in Sri Lankan private hospitals follows a clearly defined sequence involving clinical clearance, documentation, validation, billing, and payment. However, despite this structured flow, the process remains highly manual and paper dependent, requiring coordination across departments through a combination of digital systems (such as SAP) and physical document transfers.

1. **Step 1: Medical Clearance:** The Attending Physician initiates the process by granting final medical clearance, indicating that the patient is clinically fit to be discharged. This clearance is recorded in the patient's file and triggers subsequent administrative actions by ward staff.

2. **Step 2: Documentation by Ward Doctor:** Once clearance is granted, the Ward Doctor prepares two key documents, the Discharge Card and the Discharge Prescription. The discharge card summarizes treatment provided, post-discharge instructions, and any required follow-up visits. The prescription lists discharge medications to be dispensed by the pharmacy. Both documents are handed over to the Ward Nursing team for further processing.

3. **Step 3: Nursing Verification and Charge Sheet Preparation:** The Ward Nurse, typically the nurse-in-charge, conducts detailed checks and compiles three charge-related sheets:

    i. Professional Attendance and Fee Record – captures doctor visit counts and professional charges.
    ii. Ward Procedure Charge Sheet – records all standard procedures performed (e.g., physiotherapy, injections, BCG).
    iii. Surgical Consumable Items Charge Sheet – includes consumables, medical supplies, and drugs used during the stay.

Based on the discharge prescription and patient’s clinical needs, nurses also place sales orders in the system (like SAP) for any relevant items.

They then cross-verify sales orders and the Medication Prescription Administration Record (MPAR) to ensure consistency between items billed and drugs administered during the patients stay. Once verification is complete, the nurse informs the PCO that the ward-side documentation is ready.

4. **Step 4: Patient Care Officer (PCO) Validation**: The PCO acts as the bridge between clinical and administrative units. The PCO validates all discharge related entries in the systems, ensuring that billing data aligns with the patient’s clinical records. They then visit the Pharmacy physically to return any remaining drugs and deliver the physical copy of the prescription. After confirmation, the PCO notifies the Billing Department to initiate financial clearance.

5. **Step 5: Pharmacy Processing:** The Pharmacy Department receives the physical discharge prescription from the PCO and cross-checks it against system records. Pharmacists verify the medicines, confirm stock availability, and prepare the discharge medications for release. After final validation, the medications are issued to the patient, and pharmacy related charges are updated in the billing system.

6. **Step 6: Billing Verification and Bill Preparation:** The Billing Department receives the standard bill package from the PCO, which includes three ward sheets and, if applicable, the Theatre Charge Sheet. Billing officers verify all submitted documents, ensuring that the Professional Attendance and Fee Record include doctor charges. If these are missing, billing requests the relevant details from the ward or attending doctor.

Once all sheets are verified, billing staff check laboratory reports, imaging, and drug charges to consolidate a final comprehensive bill. A patient copy is printed for payment processing.

7. **Step 7: Payment and Settlement:** At this stage, the process separates depending on the payment method:

    i. Cash/Card Payments: The patient or caregiver settles the bill directly at the billing counter. A receipt is issued immediately.

    ii. Insurance Payments: Billing staff prepare and email a claim packet to the insurer, including the summary bill, detailed breakdown, diagnosis card, and claim form.

If the insurer requests additional information, the billing team provides the required documentation via email. Once approval is received, billing finalizes the settlement. If the insurer partially covers the amount or rejects the claim, the patient is contacted to settle the balance.

8. **Step 8: Final Clearance and Document Release:** After full payment is confirmed the Billing Department informs the Ward that payments are complete. The ward then releases

the patient's final discharge documents including the discharge card, prescriptions, and final bill marking the official completion of the discharge process.

Figure 11 illustrates the overall process flow, which begins once the attending physician gives the final medical clearance for the patient to be discharged.

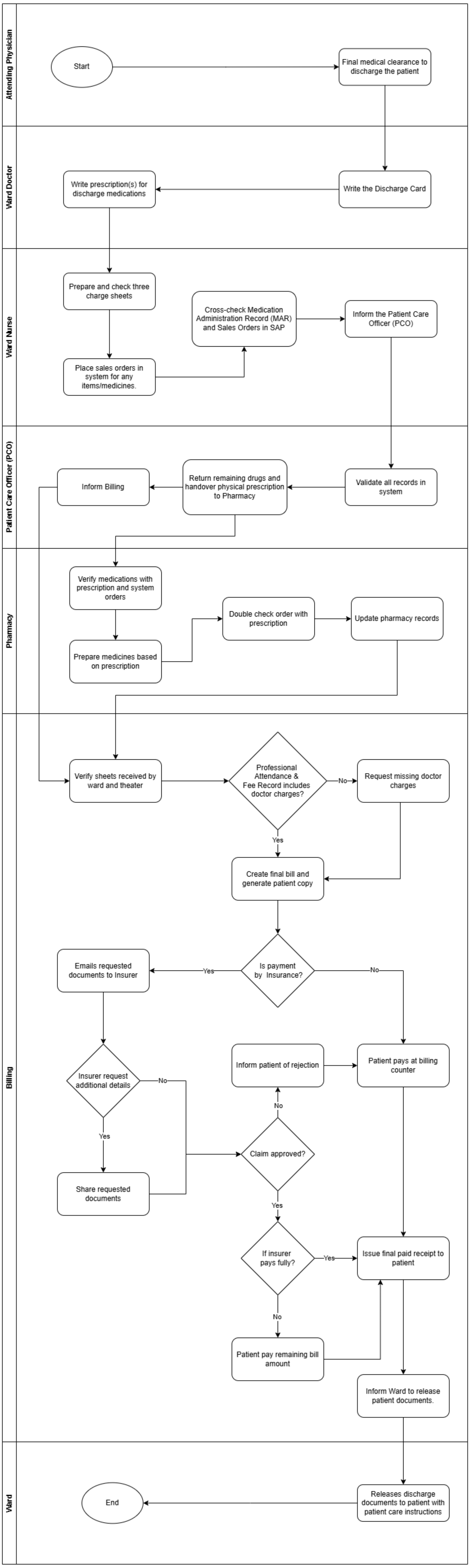

Figure 9:Current Patient Discharge Workflow in Private Hospitals (Source: Staff Interviews)

Overall, staff consistently described the workflow as sequential, verification-heavy, and dependent on physical handovers. While digital systems support record keeping, they do not provide end-to-end process organisation or real-time visibility across departments. As a result, delays collect at transition points rather than within individual tasks. These characteristics strongly indicate opportunities for RPA to function as a coordination and validation layer within future process redesigns.

4.2.3. Main Causes of Delay

Staff interviews revealed that discharge delays are not caused by a single bottleneck but comes from multiple interdependent manual steps across departments. The most significant causes of delay are summarised below.

1. Delays in Doctor Input and Documentation

One of the most frequently mentioned bottlenecks was the delay in completing the discharge card and professional charge entries by doctors.
In many cases, ward doctors are engaged in multiple patient rounds or emergency cases at the time of discharge, resulting in postponed documentation. Since billing cannot proceed without the doctor's fee record or discharge summary, this step creates a critical dependency that holds up the entire process.

2. Manual Preparation and Validation by Nursing Staff

The nursing team, especially the nurse-in-charge, plays a central role in validating charge sheets, preparing system entries, and ensuring all consumables and procedures are billed correctly.
For patients with longer stays or complex cases, this verification process becomes time-consuming, as nurses must cross-check between physical files and system entries to prevent errors. The manual nature of this validation introduces delay, particularly during high ward occupancy periods.

3. Patient Care Officer (PCO) Cross-Checking Requirements

PCOs are required to revalidate all ward and pharmacy entries in the system before informing billing. In cases with lengthy stays or multiple service entries, this step can take significant time, especially when discrepancies are found between the system data and the physical charge sheets. PCOs often spend additional time reconciling these mismatches manually, further extending discharge preparation time.

4. Physical Prescription Dependency at the Pharmacy

Pharmacy staff reported that discharge medicines can only be released once the physical prescription is received from the ward, as electronic copies are not accepted for dispensing.

Since the PCO must physically carry this document from the ward to the pharmacy, any delay in transfer particularly in larger hospital layouts directly postpones the patient's ability to collect discharge medications.

5. Insurance Company Response Times

For patients with insurance-based payments, the final discharge is dependent on insurer approval, which can take several hours.
Billing staff noted that after the claim packet is emailed, insurers sometimes request additional information or clarifications, requiring back-and-forth communication before payment can be confirmed.
This step is outside the hospital's control, yet it significantly impacts discharge timelines, particularly during weekends or after business hours.

6. System and Documentation Complexity

Multiple staff members highlighted the duplication of work between physical files and system entries.
Each department nursing, PCO, and billing performs its own validation, often repeating similar checks to ensure accuracy.
In the absence of a unified workflow or automated validation mechanism, these redundant verifications contribute to cumulative delays and potential data inconsistencies.

7. Time Required for Typing and Finalizing the Discharge Card

Ward doctors mentioned that typing the discharge summary which includes treatment history, follow-up instructions, and medications also consumes considerable time, especially for lengthy inpatient cases. This task cannot easily be delegated or automated in the current setup, and delays in finalizing the discharge card cascade into subsequent steps such as nursing verification and billing.

Table 2 summarizes the identified causes of delay and their responsible departments.

Cause of Delay	**Responsible Department(s)**	**Nature of Delay**
Delay in doctor charge entry / discharge card	Ward Doctor	Dependent on doctor rounds or emergencies
Manual charge validation	Ward Nursing	Time-consuming cross-checks in system and files
Multiple system validations	PCO / Billing	Duplicate checking increases processing time
Physical prescription transfer	Pharmacy / PCO	Dependent on physical document handover
Insurance claim processing	Billing / Insurer	Delays due to insurer response time

Complex or lengthy documentation	Ward Doctor / Nursing	Longer preparation for extended patient stays
Repetitive system entries	Multiple departments	No centralized automated data flow

Table 1: Summary of Key Delay Factors

Across all departments, staff highlighted duplication of work as a key inefficiency. Similar validations are performed independently by nursing, PCO, pharmacy, and billing teams, reflecting the absence of a centralised workflow or automated validation mechanism.

4.2.4. Awareness and Attitude Toward Automation

Findings from the staff questionnaire showed that overall awareness of RPA among hospital staff was very limited. Most respondents across all departments including nursing, pharmacy, billing, and administrative units stated that they had not previously heard of RPA or were uncertain about its application in hospital settings. This lack of exposure exhibits the current stage of technological adoption within Sri Lankan private healthcare, where automation initiatives remain largely focused on system based billing.

Because of this knowledge gap, many participants found it difficult to identify specific processes that could be automated. Their responses focused on general inefficiencies such as manual data entry, billing reconciliation, or interdepartmental communication, rather than specific automation use cases.

Despite this unfamiliarity, the staff showed a positive and open attitude toward adopting automation technologies once the concept was explained. Most expressed willingness to use RPA or similar tools provided that appropriate training, technical support, and clear implementation guidance were available. Several staff members also emphasized that automation should not replace human decision-making in clinical tasks but could help reduce repetitive administrative workload, improve accuracy, and shorten discharge time.

Overall, the findings suggest that while awareness is low, acceptance potential is high. The workforce shows readiness to embrace automation if hospitals invest in awareness programs, user training, and gradual technology integration aligned with existing workflows. This indicates a favourable organizational setting for introducing RPA solutions, particularly in administrative and billing functions.

4.2.5. Summary of Staff Insights

Overall, the staff interviews provided a comprehensive understanding of how the patient discharge process operates in practice within Sri Lankan private hospitals. The findings revealed that while the existing workflow is functionally structured, it remains heavily manual, department-dependent, and documentation intensive, which contributes to frequent inefficiencies and extended discharge times.

Staff across departments confirmed that the process involves multiple handoffs from ward to PCO, pharmacy, and billing each requiring verification, system entries, and manual cross-checking of documents. The Ward Nursing and Billing teams bear the greatest workload, as they are responsible for validating charge sheets, sales orders, and doctor fees before billing can be finalized. In contrast, Pharmacy and PCO roles are constrained by procedural dependencies such as the need for physical prescriptions and inter-system data validation.

Several recurring causes of delay were identified, including late completion of doctor charges, time-consuming data entry and verification by nursing staff, physical prescription transfers, and slow insurer response times. These challenges come largely from decentralised digital systems, lack of automation, and reliance on individual staff availability rather than real-time data integration.

In terms of technology awareness, most staff had no prior exposure to RPA. However, the overall attitude toward automation was highly positive, with participants expressing willingness to adopt such technologies if proper training and support are provided. Staff acknowledged that automation could ease repetitive administrative work, reduce manual validation, and accelerate billing and discharge approval processes.

In summary, the staff insights highlight that:

- The current discharge process is structured but inefficient, with heavy reliance on manual data handling.
- Cross-departmental coordination is a major source of delay.
- System fragmentation (between SAP and paper documents) causes redundancy and re-work.
- Staff demonstrate low awareness but high readiness toward automation adoption.
- Administrative functions, particularly billing, PCO validation, and pharmacy communication, present the greatest potential for RPA integration.

These insights provide the foundation for identifying automation opportunities and designing a context sensitive RPA solution suitable for Sri Lankan private hospitals, discussed in the subsequent sections.

4.3. Comparative Analysis

This comparative analysis synthesises findings from the literature review with evidence gathered from patient surveys and staff interviews to evaluate the discharge processes in Sri Lankan private hospitals relative to global healthcare automation practices. The analysis proves that the types of discharge challenges observed in Sri Lanka are consistent with those reported internationally, while the extent of technological advancement and process integration varies significantly.

International literature from healthcare systems in the United Kingdom, Australia, and the United States consistently reports patient discharge as a complex, inter departmental process subjected to delays due to documentation dependencies, billing validation, and insurance coordination. However, these systems have increasingly solved such challenges through the adoption of Robotic Process Automation, integrated Electronic Health Records, and process orchestration tools. In these settings, automation is not applied to clinical decision making but to administrative coordination which reduced manual effort, eliminating repeated validations, and improving process transparency.

In contrast, findings from this study shows that Sri Lankan private hospitals operate at a lower level of digital and process automation maturity. While transactional systems such as SAP and Hospital Information Systems are in use, they function as isolated data sources rather than as integrated system. As a result, discharge activities continue to rely heavily on paper based documentation, physical handovers, and repeated human verification across departments. This fragmented digital environment explains why discharge delays in Sri Lanka are evident despite the presence of basic information systems.

Globally implemented RPA based discharge flows commonly reveal four core characteristics: automated validation of billing and insurance documentation; real-time data synchronisation between clinical and financial systems; electronic transmission of prescriptions and discharge summaries; and automated insurer communication with escalation and tracking mechanisms. These features collectively reduce dependency on individual staff availability and transform discharge into a parallel, rather than sequential, process.

Findings from Sri Lankan hospitals reveal reliance on individual expertise, particularly among senior nursing staff and ward doctors; limited awareness of automation technologies; and absence of system driven coordination mechanisms. Insurance related discharges remain particularly vulnerable to delay due to the lack of automated claim submission, monitoring, or follow-up. These findings suggest that inefficiencies are not caused by resistance to change but by structural and technological limitations within existing systems.

Importantly, the comparative analysis also identifies a critical point of alignment between global best practices and the Sri Lankan context which is stakeholder readiness. Both patient and staff data display strong acceptance of digitalisation and automation, reflecting findings from early automation phases in international healthcare settings. This suggests that Sri Lankan private hospitals organisationally and culturally are not against automation initiatives but instead at an earlier stage of the automation adoption lifecycle.

Therefore, when evaluated against global benchmarks, Sri Lankan private hospitals can be characterised as being at an introductory stage of administrative automation maturity, where foundational systems exist but lack coordination, interoperability, and automation layers. This gap represents not a barrier but an opportunity. A phased introduction of RPA focused initially on non-clinical, high-volume administrative tasks such as billing reconciliation, pharmacy coordination, and insurance claim processing offers a realistic and context sensitive pathway toward discharge process optimisation.

In summary, this comparative analysis demonstrates that while discharge inefficiencies in Sri Lankan private hospitals are structurally similar to those experienced globally, the absence of digital maturity differentiates the local context. The merging of documented inefficiencies, stakeholder openness, and proven international success positions RPA as a suitable and timely solution for improving discharge efficiency, transparency, and patient satisfaction within the Sri Lankan private healthcare sector.

4.4. Summary of Findings

This chapter analysed and interpreted findings from both patient and staff perspectives to examine the effectiveness of the current patient discharge process in Sri Lankan private hospitals and to assess the suitability of automation as an improvement mechanism. By analysing survey data, staff interviews, and comparative insights from international literature, the chapter provides a detailed understanding of existing inefficiencies and organisational readiness for change.

From the patient perspective, the findings reveal that discharge delays are common, with a significant proportion of respondents experiencing waiting times exceeding three hours after receiving medical clearance. Patients consistently identified administrative activities particularly billing, pharmacy processing, and interdepartmental approvals as the primary sources of delay, rather than clinical readiness. Despite these inefficiencies, patients generally expressed positive perceptions of staff professionalism and demonstrated strong support for digitalisation and automation as a means of improving transparency, speed, and overall service quality.

Staff findings further broaden these experiences by revealing a discharge workflow that is highly sequential, documentation intensive, and dependent on manual coordination across multiple departments. Bottlenecks were consistently associated with delays in doctor documentation, time consuming charge validation by nursing staff, physical prescription dependencies at the pharmacy, and insurer response times for insurance patients. These issues are caused by fragmented system usage and repeated manual verification, leading to duplication of effort and extended discharge time. Although staff awareness of RPA and automation concepts was low, attitudes toward automation were positive once the concept was explained, indicating high acceptance potential if appropriate training and support are provided.

The comparative analysis confirmed that the challenges observed in Sri Lankan private hospitals closely resemble those documented in international healthcare systems prior to automation adoption. However, unlike technologically mature environments where RPA and integrated workflows have reduced administrative delays, Sri Lankan hospitals remain held back by limited interoperability and absence of process orchestration. This gap highlights not a resistance to automation, but a lack of technological maturity.

Collectively, the findings determine that the current discharge process is operationally functional but systemically inefficient, with delays driven primarily by administrative. Both patients and staff experience the consequences of these inefficiencies through persistent waiting times, repetitive data checks, and limited process visibility. At the same time, strong stakeholder openness toward automation and clear alignment with global pre automation challenges indicate a favourable context for introducing targeted, non-clinical automation initiatives.

In summary, the findings establish a clear reasoning for the development of an RPA based discharge workflow tailored to the Sri Lankan private healthcare context. By addressing administrative coordination gaps, reducing manual verification, and improving interdepartmental information flow, such a framework has the potential to enhance operational efficiency and patient experience. These insights directly guide the design of the proposed solution presented in Chapter 5.

5. Proposed Solution

5.1. Introduction

This section provides a logical and conceptual solution designed to address the inefficiencies identified in the current patient discharge process within Sri Lankan private hospitals.

The findings discussed in Chapter 4 revealed that the existing process is heavily manual, time-consuming, and highly dependent on human coordination across departments such as Nursing, Pharmacy, Billing, and Insurance. Therefore, this chapter focus on how RPA based discharge framework could adopt to optimize the workflow. It is important to note that this solution is theoretical in nature, serving as a conceptual framework than a system implementation and the purpose is to prove how automation could enhance discharge efficiency and accuracy if hospitals choose to adopt it in the future.

5.2. Rationale for Automation

The rationale for proposing automation comes from the recurring inefficiencies and delays highlighted by both patients and hospital staff during data collection.

The current workflow requires manual effort for data validation, document verification, and interdepartmental coordination, which contributes to prolonged discharge times and administrative workload.

Key justifications for automation include:

- Multiple manual validations across Nursing, PCO, and Billing departments.
- Dependence on physical documentation such as prescriptions and charge sheets.
- Repetitive and error-prone data entry between systems, and manual records.
- Delayed insurer communication and follow-ups due to lack of automation.
- Staff exhaustion and resource strain during peak admission or discharge periods.

RPA can directly address these issues by automating repetitive, rules-based administrative activities while leaving clinical decision-making to human professionals.

As seen in global healthcare examples, RPA tools are capable of validating data, generating bills, notifying departments, and submitting insurance claims with minimal human intervention.
Therefore, introducing RPA at a conceptual level offers a feasible pathway for Sri Lankan hospitals to enhance their discharge efficiency and service quality.

These observations indicate that the discharge process is particularly suitable for RPA, as it involves high volume, rule based administrative activities.

5.3. Proposed RPA-Integrated Discharge Framework

The proposed framework integrates automation bots at specific stages of the discharge process where routine, repetitive, and document-heavy tasks occur. The system is designed to work alongside existing digital system platform, serving as an additional automation layer that bridges interdepartmental workflows.

5.3.1. Automation Points

Automation Bot / Function	**Purpose / Role**	**Automated Activities**
RPA Bot 1 – Data Validation Bot	Streamline nursing and PCO checks	Cross verifies discharge data, system sales orders, and charge sheets to ensure accuracy before billing.
RPA Bot 2 – Notification Bot	Improve communication flow	Sends automated notifications to Pharmacy and Billing once PCO validation is complete.
RPA Bot 3 – Pharmacy Bot	Streamline prescription and stock handling	Validates prescription details, updates pharmacy inventory, and informs Billing when medicines are released.
RPA Bot 4 – Billing Bot	Consolidate multi department billing	Aggregates charges from Ward, Theatre, and Pharmacy, verify doctor fees and prepares the preliminary bill in system.
RPA Bot 5 – Insurance Bot	Automate insurance correspondence	Prepares and emails claim packets to insurers, monitors responses, and updates Billing on claim status.

Table 2: Automation Options within the Discharge Flow

5.3.2. Proposed Workflow

In the redesigned process:

- The Attending Physician and Ward Doctor continue to handle clinical duties such as clearance and discharge documentation.
- Once discharge data is entered into system, RPA bots automatically trigger validations, notifications, and billing updates.
- Communication between PCO, Pharmacy, and Billing occurs in real time through automated alerts.
- Insurance processing and claim tracking are automated, minimizing manual delays and repetitive email exchanges.

This framework ensures that administrative coordination becomes digital, synchronized, and data driven, significantly reducing time and error rates.

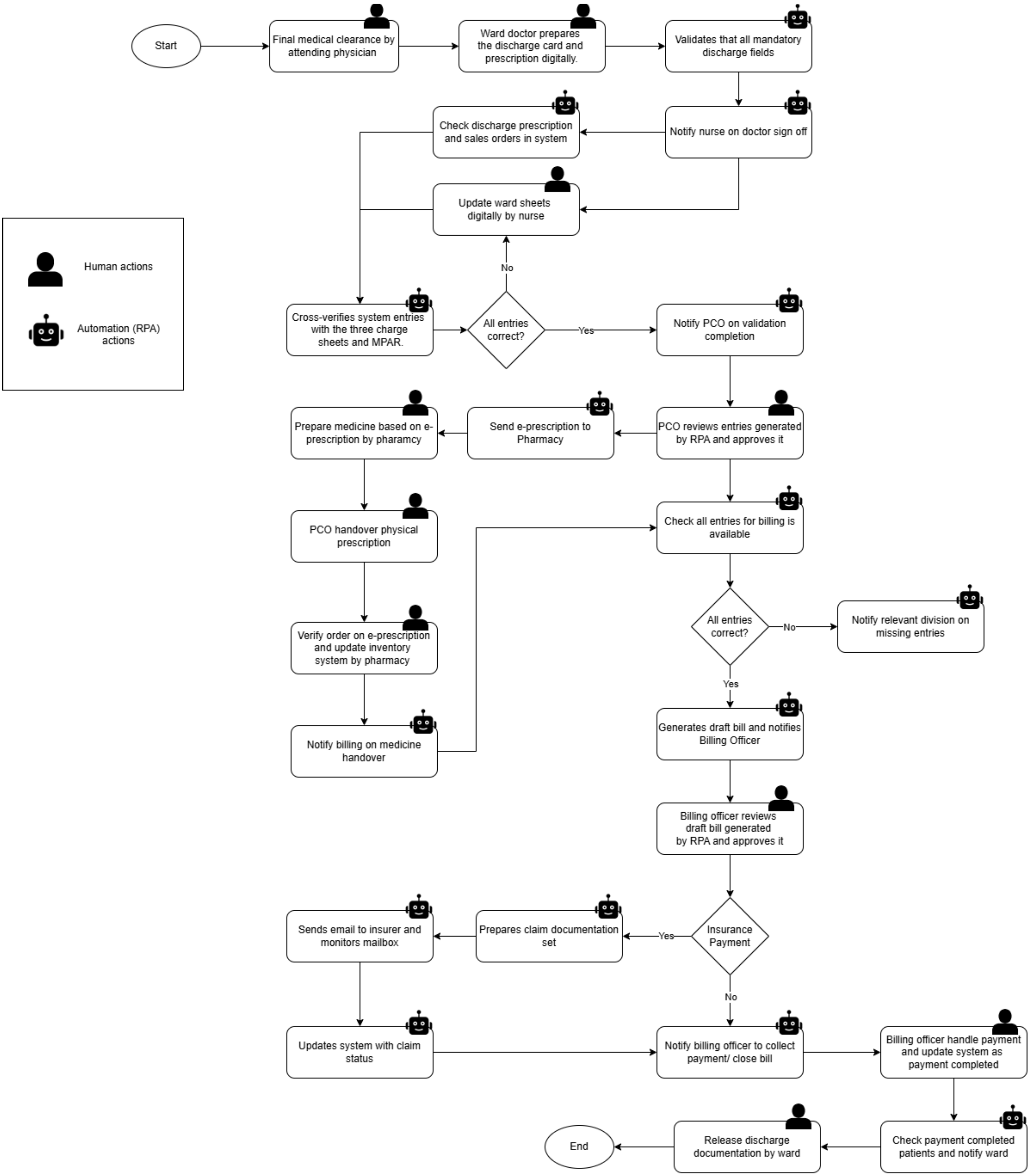

Figure 10: Proposed RPA-Integrated Discharge Workflow (Conceptual Model)

5.4. System Architecture Overview

The conceptual system architecture demonstrates how RPA can function as a middleware layer that connects existing hospital systems without replacing them.

System Components:

- Core Systems: Hospital ERP system (example: SAP), HIS systems, email servers and any other system used during discharge flow.
- Automation Layer: RPA software such as UiPath, Blue Prism, or Microsoft Power Automate.
- Data Inputs: Charge sheets, discharge summaries, and prescriptions (**scanned or electronic**).
- Outputs: Validated records, automated notifications, insurance claim submissions, and status dashboards.

Process Flow:

- Data entered by Nursing or Billing triggers RPA workflows.
- Bots read and validate records, perform assigned actions, and update the corresponding system.
- Each bot logs its actions for traceability, ensuring compliance and audit readiness.
- Administrators can monitor discharge status and performance metrics via a centralized dashboard.

The modular structure allows hospitals to implement automation incrementally, starting with billing or insurance tasks before scaling to other functions.

5.5. Implementation Roadmap

Although this study proposes a logical model, the following phased roadmap illustrates how hospitals could realistically implement RPA over time.

Phase	Focus Area	Key Actions	Expected Outcome
Phase 1 – Pilot Automation	Billing Validation	Deploy the Data Validation and Billing Bots in one ward or department.	Faster billing and reduced manual corrections.
Phase 2 – Department Expansion	PCO and Pharmacy Coordination	Introduce Notification and Pharmacy Bots.	Real time communication and faster medication clearance.

Phase 3 – Insurance Handling	Claims Automation	Implement Insurance Bot for claim submissions and response tracking.	Quicker insurer approval and fewer communication delays.
Phase 4 – Organization-Wide Rollout	Full Integration	Connect all RPA bots under a unified monitoring dashboard.	Complete discharge automation and real-time oversight.

Table 3: Implementation Roadmap

5.6. RPA Platform Considerations

This study does not evaluate, compare, or promote any specific RPA vendor. Instead, it adopts a context based and unbiased perspective, recognising that the suitability of an RPA platform depends on organisational characteristics, existing IT infrastructure, process complexity, and resource availability within private hospitals.

Private hospitals in Sri Lanka exhibit varying levels of digital maturity, system integration, and financial capacity. As a result, a single RPA platform may not be commonly appropriate across all hospitals. For example, hospitals operating within a Microsoft based environment may benefit from RPA solutions that offer native integration with its tools and require minimal additional infrastructure. On the other hand, hospitals managing complex, high volume workflows across multiple legacy systems may require more advanced automation platforms that provide stronger orchestration, governance, and scalability capabilities.

Accordingly, RPA platform selection should be guided by background fit rather with emphasis placed on alignment with existing systems, ease of adoption, compliance requirements, and long-term sustainability. The examples presented below are advisory and intended to support informed decision making rather than vendor endorsement.

Hospital Context	**Typical Characteristics**	**Illustrative RPA Platform Examples**	**Suitability Rationale**
Microsoft-centric IT environment	Extensive use of Microsoft 365, Outlook, Excel, SharePoint	Microsoft Power Automate (Microsoft, 2025)	Smooth integration with existing tools, lower entry cost, suitable for basic administrative automation
Large private hospital groups	High transaction volumes, multiple departments, complex workflows	UiPath (UiPath, 2025)	Advanced orchestration, scalability, and governance capabilities suitable for enterprise-level automation
Highly regulated environments	Strong audit, compliance, and security requirements	Blue Prism (SS&C Blue Prism, 2025)	Robust governance, role based access control, and auditability

Budget-constrained hospitals	Limited IT budgets, early stage digital maturity	Power Automate (Microsoft, 2025)/ Open-source RPA tools	Lower licensing costs and phased automation potential
Paperheavy workflows transitioning to digital	Reliance on scanned documents and manual forms	UiPath (UiPath, 2025) / Automation Anywhere (Automation Anywhere, Inc., 2025)	Strong OCR and document processing capabilities to support gradual digitisation

Table 4: RPA Platform Suitability Based on Hospital Context

5.7. Expected Benefits

If adopted, the proposed framework could bring multiple short-term and long-term benefits for private hospitals:

Operational Benefits

- Significant reduction in discharge processing time.
- Improved coordination between departments through automated notifications.
- Decrease in manual data entry errors and redundant checks.

Financial Benefits

- Shorter billing cycles and quicker insurer reimbursements.
- Reduced administrative costs due to workload optimization.
- Clearer visibility of revenue and claim status.

Patient and Staff Benefits

- Shorter waiting periods for discharge and improved patient satisfaction.
- Reduced workload and stress for administrative and nursing staff.
- Better transparency in discharge progress for all stakeholders.

Organizational Benefits

- Standardized discharge workflow across all departments.
- Enhanced data integrity and compliance.
- Stronger institutional image as a digitally advanced healthcare provider.

5.8. Risks and Mitigation Strategies

Potential Risk	**Impact**	**Mitigation Strategy**
Integration Challenges	Difficulty linking RPA with existing systems.	Conduct pilot integration testing and use industry know RPA tools with industry level connectors available. Example: UiPath, Power Automate has SAP system connectors within the tool.
Staff Resistance	Reluctance due to job security fears or unfamiliarity.	Awareness programs and hands-on training emphasizing support, not replacement.
Data Privacy Concerns	Risk of unauthorized access or data leakage.	Implement strict access control, audit trails, and encrypted connections.
Process Variations	Differences between wards or departments.	Standardize discharge workflows before automation.
Financial Investment	Initial cost of software and consultancy.	Begin with small-scale pilots and demonstrate ROI to justify expansion.

Table 5: Risks and Mitigation

5.9. Summary

This chapter proposed a conceptual RPA based discharge solution that aims to address the inefficiencies identified in the current discharge process. The framework focuses on automating repetitive, administrative tasks rather than clinical activities, ensuring that human judgment remains central while improving efficiency and accuracy.

By introducing automation in data validation, interdepartmental communication, billing consolidation, and insurance handling, hospitals can significantly reduce discharge delays and improve patient experience. Although this research does not involve technical implementation, the proposed solution serves as a logical blueprint that Sri Lankan private hospitals can adapt when pursuing digital transformation.

The next chapter examines the feasibility of implementing the proposed solution, with particular emphasis on process refinement requirements, organisational readiness, and ethical, legal, and data protection considerations relevant to RPA adoption in Sri Lankan healthcare.

6. Discussion

6.1. Preconditions for Implementing the Proposed Solution

Chapter 5 presented a conceptual RPA enabled discharge framework designed to address the inefficiencies identified in the existing patient discharge process. However, the findings of this research indicate that the successful implementation of the proposed solution is depending upon prior refinement of current discharge workflows. The existing process within Sri Lankan private hospitals contains fragmented task ownership, delayed clinical documentation, and hybrid paper and digital practices, which collectively limit the readiness for automation.

Robotic Process Automation is fundamentally dependent on structured, rule-based, and digitally triggered processes. In the absence of real-time data entry particularly at the clinical documentation stage, the automation points outlined in the proposed framework cannot function as intended. As such, the proposed solution cannot be implemented effectively in its current operational context without addressing these foundational process limitations.

This research therefore argues that process refinement must be done first before any automation adoption. Key preconditions include the standardisation of discharge workflows across departments, the enforcement of digital first documentation at the point of care, and the clarification of interdepartmental roles and responsibilities. These refinements are necessary to ensure that accurate and timely data is available to support automated validation, notification, billing, and insurance workflows.

By recognising these prerequisites, this chapter provides a bridge between the conceptual solution presented in Chapter 5 and its practical feasibility. This positioning reinforces the view that RPA should be applied for the optimised processes rather than as a substitute for process improvement.

6.2. Process Re-engineering Requirements

Expanding on the preconditions identified, this research argues that effective implementation of the proposed RPA enabled discharge framework requires targeted refinement of documentation practices and interdepartmental workflows. Central to this improvement is the digitisation of ward-level documentation and clinical discharge records, which currently constitute a significant bottleneck in the discharge process.

For automation to operate reliably, all core discharge-related documents including ward sheets, discharge summaries, prescriptions, and charge-related entries must be generated digitally by medical staff at the point of care. Digital-first documentation allows automated systems to perform rule-based verification and cross-checking without continuous human intervention. In the absence of such digitisation, discharge workflows remain dependent on manual validation, undermining the efficiency gains expected from automation.

Once discharge documentation is digitally completed and validated, the prescription process can be streamlined through the use of electronic prescriptions. E-prescriptions may be transmitted automatically to the pharmacy, allowing medications to be prepared in advance in accordance with established clinical guidelines. While physical prescriptions may still be required at the point of medicine collection to comply with regulatory or operational practices, automation enables preparatory activities to occur without delay, thereby reducing overall discharge turnaround time.

Similarly, insurance related workflows benefit significantly from process re-engineering. When discharge data, billing entries, and clinical documentation are available in structured digital formats, RPA bots can automatically compile insurance claim documentation sets, eliminating the need for repetitive manual preparation. Automated communication with insurers including claim submission, status monitoring, and follow-up correspondence can further reduce administrative workload and minimise delays associated with manual email-based coordination.

Together, these refinements reposition the discharge process as a digitally coordinated workflow rather than a sequence of isolated departmental tasks. By ensuring that documentation originates in digital form and flows seamlessly across clinical, pharmacy, billing, and insurance functions, automation can replace manual rechecking and communication activities. This transformation creates the necessary operational foundation for RPA to function as an effective enabler of efficiency, accuracy, and timely patient discharge.

6.3. RPA in Re-engineered Workflows

Robotic Process Automation has the potential to significantly improve discharge efficiency when applied to re-engineered workflows. RPA is particularly well suited for automating repetitive, rule-based administrative tasks such as billing validation, insurance coordination, discharge notifications, and pharmacy clearance processes.

However, the findings also highlight critical limitations of RPA within the current healthcare environment. RPA tools are unable to interpret unstructured handwritten clinical notes, make up for delayed documentation, or resolve uncertainties arising from inconsistent data entry practices. Therefore, RPA adoption without prior process and data standardisation is unlikely to produce sustainable improvements.

This emphasizes the view that RPA should be deployed as a supporting mechanism rather than as a standalone solution. When aligned with redesigned workflows and supported by organisational readiness initiatives, RPA can function as a performance enhancer. In the absence of such conditions, its impact remains limited.

6.4. From RPA to Intelligent Automation

While RPA offers immediate operational benefits, it represents an early stage in automation maturity. Intelligent Automation (IA), which integrates artificial intelligence, machine learning, and natural language processing with RPA, has the potential to further optimise discharge processes through predictive and adaptive capabilities.

This study proposes a phased automation roadmap. The initial phase focuses on process re-engineering and digital maturity, ensuring that discharge related data is captured in structured formats at the point of care. The second phase involves the deployment of RPA to automate rule-based discharge activities. Only after achieving stability and data consistency should healthcare organisations consider transitioning to Intelligent Automation.

Potential IA applications include predictive modelling of discharge readiness, automated discharge summary generation, and intelligent anomaly detection within billing workflows. However, such applications require high quality datasets, advanced governance mechanisms, and ethical oversight frameworks that are currently underdeveloped in the Sri Lankan healthcare context. As such, Intelligent Automation should be regarded as a long-term strategic objective rather than an immediate implementation option.

6.5. Ethical Aspects

The adoption of Robotic Process Automation within healthcare environments raises a range of ethical considerations beyond technical feasibility. This research recognises that automation initiatives must be implemented in a way that safeguards human roles, maintains accountability, and upholds ethical research and operational standards.

Job Displacement and Transformation

One of the most frequently mentioned ethical concerns associated with automation is the potential displacement of human labour through the automation of repetitive tasks. Within the healthcare context, this concern is particularly sensitive due to the critical role of human judgement and care delivery. This research adopts a balanced perspective, positioning RPA not as a replacement for healthcare professionals but as tool enhancing human capabilities.
The proposed use of RPA focuses on automating routine, administrative activities rather than clinical decision-making, thereby allowing healthcare staff to reallocate their efforts toward higher value, patient centred tasks. Therefore, the focus is placed on job transformation rather than job elimination, with automation serving to enhance productivity and reduce administrative burden rather than decreasing employment opportunities.

Transparency and Accountability

Ethical deployment of RPA requires a high degree of transparency, particularly where automated systems interact with sensitive healthcare workflows. Automated decisions and actions must remain traceable, auditable, and subject to human oversight to prevent unintended consequences. This research advocates the implementation of a "human-in-the-loop" governance model, whereby automated processes operate under human supervision. Such an approach ensures that all actions performed by bots are transparent and that accountability remains with designated human stakeholders. Maintaining audit trails and exception-handling mechanisms further supports ethical accountability and trust in automated systems.

Informed Consent

Ethical research practice requires that all primary data collection activities be conducted with informed consent. In this study, participants involved in questionnaires and surveys were fully informed of the research objectives, the voluntary nature of their participation, and their right to withdraw at any stage without consequence.

Participants were also informed about how their data would be collected, stored, and used exclusively for academic purposes. This approach ensures compliance with established ethical research standards and reinforces respect for participant autonomy and privacy.

6.6. Legal Considerations

The implementation of Robotic Process Automation within healthcare environments must operate within a clearly defined legal and regulatory framework. Given the sensitive nature of healthcare operations and patient data, automation initiatives that fail to address legal requirements may lead organisations to significant liability risks.

Data Handling and Ownership

RPA systems deployed in hospital environments may process sensitive personal and medical data, including patient identifiers, clinical records, and billing information. This research emphasises the importance of clearly defining data ownership, access rights, retention policies, and confidentiality obligations prior to automation deployment.

Automated processes must adhere to existing legal requirements governing healthcare data handling, ensuring that data is accessed strictly on a need to process basis and retained only for legally permitted durations. Any automation solution implemented basis of the suggested framework must therefore be designed in alignment with institutional data governance policies and applicable healthcare regulations, ensuring that responsibility and accountability for data usage remain clearly assigned.

Automation of Regulated Processes

Certain hospital workflows, particularly those related to billing, insurance claims, and clinical documentation, are subject to national laws, professional standards, and regulatory oversight. Automating such processes introduces legal risks if workflows are not carefully reviewed and governed.

This research highlights the necessity of implementing strong governance frameworks to oversee automated processes operating within regulated domains. Such frameworks should include process validation, regular audits, and escalation mechanisms to ensure that RPA bots function within legally defined boundaries.

Intellectual Property

Respect for intellectual property rights is a fundamental legal and ethical requirement in academic research and system design. This study ensures that all external sources, frameworks, and automation platforms referenced are appropriately cited in accordance with academic standards.
Furthermore, any third-party RPA tools or technologies discussed within the proposed solution are acknowledged in line. This approach safeguards against intellectual property violations while maintaining transparency regarding the use of proprietary technologies.

6.7. Data Protection Considerations

Data protection constitutes a critical consideration in healthcare automation due to the highly sensitive nature of patient information. The adoption of RPA must therefore be aligned with established data protection frameworks to ensure confidentiality, integrity, and lawful processing of personal data.

GDPR and Related Frameworks

This research recognises the importance of aligning automation initiatives with international and regional data protection regulations, including the General Data Protection Regulation (GDPR) and comparable healthcare data protection standards such as HIPAA. Although the study is situated within the Sri Lankan healthcare context, adherence to globally recognised frameworks provide a strong benchmark for responsible data handling.

The proposed automation approach is designed to minimise data exposure by restricting bot access to only the information required to perform specific tasks. Additionally, maintaining comprehensive audit trails ensures traceability of automated actions, supporting accountability and regulatory compliance.

Data Encryption and Access Controls

Ensuring data security is essential when introducing automation into hospital systems. This research recommends that all data transfers associated with automated workflows be protected through encryption mechanisms to prevent unauthorised access or interception.

Role-based access control (RBAC) is also identified as a key security measure, enabling system access to be restricted according to defined user and bot roles. By limiting access privileges and monitoring automated activities, healthcare organisations can reduce the risk of data breaches while maintaining operational efficiency.

7. Conclusion

This research examined the potential of RPA to enhance the patient discharge process in Sri Lankan private hospitals, addressing a crucial yet under researched challenge within the local healthcare sector. The patient discharge process is naturally complex, involving coordination among clinical, administrative, financial, and external stakeholders. Inefficiencies in this process have been shown to negatively affect patient satisfaction, bed utilisation, and overall hospital performance. Despite the global adoption of RPA in healthcare, its application within Sri Lanka remains limited in both academic literature and practice. This study aims to bridge that gap by evaluating how RPA could be strategically and responsibly applied within the Sri Lankan private healthcare. A qualitative research approach was adopted, combining a systematic literature review, analysis of existing discharge workflows, and primary data collected through staff questionnaires and patient surveys. This approach allowed an understanding of both the technical potential of automation and organisational readiness for adoption.

The findings indicate that discharge inefficiencies in Sri Lankan private hospitals are primarily driven by manual documentation, fragmented information flows, repeated data entry, and reliance on paper-based approvals. While these challenges are consistent internationally, they are further intensified locally by legacy systems, limited digital integration, and uneven IT maturity. In response, the study proposed an RPA based discharge framework that clearly separates tasks suitable for automation from those requiring clinical judgment or human oversight. The framework demonstrates how RPA can be layered onto re-engineered workflows to reduce delays, improve data accuracy, and enhance inter-departmental coordination without compromising clinical independence. A key contribution of this research is in its context aware evaluation of RPA adoption. Rather than promoting a single automation platform, the study presented a comparative, situational perspective on RPA tools and supporting technologies, highlighting how organisational context such as existing IT ecosystems, process complexity, and resource constraints should guide tool selection. Academically, this research extends RPA literature into a South Asian healthcare setting. Practically, it provides healthcare administrators with a framework that can be adapted to their specific operational environments.

From an organisational perspective, the study offers practical guidance for private hospitals considering automation. The proposed framework and implementation roadmap support incremental adoption, beginning with process standardisation and digital readiness, followed by selective automation of high-impact administrative tasks. The research also highlights that RPA should complement, rather than replace, human roles, placing automation as a means of reducing administrative burden and allowing staff to focus on higher-value, patient-centred activities. The inclusion of ethical, legal, and data protection considerations further improves the relevance of the study in highly regulated healthcare environments.

Several limitations should be acknowledged. The study relied on qualitative data from a limited sample of private hospitals, which may affect generalisability. In addition, the absence of a live RPA implementation means that performance improvements were assumed rather than measured. These limitations point to opportunities for future research. Future studies could conduct in-depth case studies of RPA implementation in Sri Lankan hospitals to quantitatively measure outcomes such as discharge time reduction, cost savings, and patient satisfaction. Further research could also explore the integration of RPA with intelligent automation techniques, including process mining and predictive analytics, as hospitals progress toward higher digital maturity. Another future work would be extending the analysis to public healthcare institutions that would provide valuable comparative insights.

In conclusion, this study shows that RPA has significant potential to improve patient discharge efficiency in Sri Lankan private hospitals when implemented through a structured, context aware, and ethically grounded approach. By aligning automation initiatives with organisational readiness, the research provides a strong foundation for informed decision-making and contributes meaningfully to the digital transformation in Sri Lanka healthcare.

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Appendices

Appendix A – Literature Review Summary

Author	Title	Key Takeaway
Kumar, Jatin	A Study of the Causes of Delay in Patient Discharge Process in a Large Multi-Speciality Hospital with Recommendations to Improve the Turn around Time	• Delays in the discharge process are recognized as major bottlenecks that increase costs, consume beds, and frustrate patients • The average time taken for discharge of cash patients is 3 h and 57 min and for insurance patients is 5 h and 9 min
Paula R. Patel; Samuel Bechmann	Discharge Planning	• Discharge often starts too late, particularly for patients with chronic or complex conditions, which increases the risk of complications or readmissions.
Michael Emes, Stella Smith, Suzanne Ward, Alan Smith	Improving the patient discharge process: implementing actions derived from a soft systems methodology study	• Patients are typically discharged once declared medically stable, but actual discharge often faces delays due to administrative and coordination issues. • The process involves medical clearance, care planning, social care involvement, and final paperwork each step requires coordination between multiple professionals and departments.
Saurabh Sharma, Ravi Pratap Singh, Archita Kansal Tiwari, Pawan Parashar	Analysis of Time Taken for the Discharge Process and its Determinants in a Tertiary Care Teaching Hospital	• Mean discharge duration: 372 minutes (approx. 6.2 hours) • Longest delay is in Preparation of discharge summary, followed by the clearance of files from various departments
Umelo FU, Onuchukwu K, Opadele MM	Improving Efficiency Of Patients Discharge Process In A Private Multi-Speciality Hospital	• There was a significant reduction in the average discharge time recorded from 469.80 minutes (7.83 hours) during the first 6 months of the pre-intervention phase to 262.80 minutes (4.38 hours) across the entire post-intervention phase indicating a 44.08% reduction.
Dr Asiri. B. Dissanayaka	Novel Strategies to Bring Private Hospital Industry in Sri Lanka to the Next Level	• The private health sector comprises more than 141 hospitals that provide in-patient services, catering to over 135,000 admissions annually

IBM	What is robotic process automation (RPA)?	• Robotic Process Automation (RPA) is software that mimics human actions to automate repetitive office tasks like data entry, form filling, and moving files. • Uses rule-based automation to handle high-volume business processes without altering underlying systems. • Next evolution of RPA: combines RPA with AI elements like machine learning, NLP, and computer vision.
UiPath	Intelligent automation	• Intelligent automation (IA) combines Robotic Process Automation (RPA) with Artificial Intelligence (AI) and Machine Learning (ML) to automate complex, end-to-end business workflows. • It is an evolution of traditional RPA, enabling automation of tasks that require cognitive abilities like document understanding and unstructured data processing.
UiPath	Use Cases for Healthcare Automation: Revenue Cycle Management	• RPA can automate tasks like scheduling, billing, claims, and audits with speed and accuracy.
Automation Anywhere	St. John of God Health Care processes $1B AUD with RPA, reduces operational costs by twenty%	• Automation Anywhere's RPA platform has helped St John of God Health Care process close to $1B Australian dollars in billing and save 25,000 hours annually, resulting in a 20% reduction in operational costs.
Automation Anywhere	How Can Healthcare Organizations Use Intelligent Automation?	• RPA bots can be used in post care coordination and follow ups.
Cleveland Clinic	Nursing Teams Leverage Automation to Improve Workflows, Devote More Time to Patient Care	• Between 45% and 52% of monthly discharge reviews are automated in the Cleveland Clinic
James Andrew, Alexander Imogen	Robotic Process Automation in Medical Record Management: A Cost-Benefit Analysis	• After RPA implementation routine documentation tasks (e.g., discharge summaries, billing, insurance verification) were completed up to 70% faster. • Error rates dropped from 5% to under 1%, significantly improving data integrity and clinical decision-making.

Prashant Nimkar, Deepika Kanyal, Shantanu R Sabale	Increasing Trends of Artificial Intelligence With Robotic Process Automation in Health Care: A Narrative Review	• In RPA implementation withing healthcare challenges such as data privacy and security concerns, high implementation costs, and regulatory and ethical considerations arises.
Redwood Software	Navigating through the limitations of robotic process automation (RPA)	• RPA excels at rule-based, structured tasks but struggles with unstructured tasks requiring human creativity, empathy, or intuition. • Scalability is a major challenge, as RPA solutions effective for small operations may not adapt well to growing enterprise demands. • Legacy system integration is problematic, with UI changes and lack of APIs causing frequent disruptions and requiring reconfiguration.
Patrick Kraus, Elias Fißler,Dennis Schlegel	A typology of challenges in the context of robotic process automation implementation projects	• RPA implementation hurdles are not just technical; they span a typology of process, technical, resource, psychological, and coordinative dimensions. • Automating unoptimized or undocumented workflows is a primary cause of project failure; process stabilization must occur before building the robot. • Psychological resistance (employee fear of job loss) and poor IT-business alignment are typically more disruptive than software limitations.
Aguirre Santiago, Rodriguez Alejandro	Automation of a Business Process Using Robotic Process Automation (RPA): A Case Study	• RPA is most effective for routine, rule-based tasks that involve structured data and predictable, deterministic outcomes. • While RPA leads to significant productivity gains, this specific study found that automation does not always result in a reduction of overall process time. • The technology is viable beyond simple back-office work, successfully handling processes with customer-facing activities in a BPO environment.
Chaturvedi Ritesh, Sharma Saloni	Robotic Process Automation (RPA) in Healthcare:	• Automating routine tasks like patient registration and billing reduces

	Transforming Revenue Cycle Operations	processing time by 30% to 40%, allowing staff to focus on complex patient care. • Organizations implementing RPA in claims management see an average 20% improvement in revenue due to faster cash flow and significantly lower claim denial rates. • Success is often limited by the complexity of integrating with legacy IT systems and the high priority required for patient data security and privacy compliance.
Deo Niyati, Anjankar Ashish	Artificial Intelligence With Robotics in Healthcare: A Narrative Review of Its Viability in India	• AI and robotics can automate routine tasks to fill the critical gap left by India's lack of skilled healthcare professionals. • Technology significantly accelerates drug discovery and can predict epidemic outbreaks before they spread. • High implementation costs and concerns over losing the "humanitarian aspect" of care remain major obstacles for adoption in India.
Nimkar Prashant, Kanyal Deepika, Sabale Shantanu R.	Increasing Trends of Artificial Intelligence With Robotic Process Automation in Health Care: A Narrative Review	• : Combining AI with RPA moves healthcare from simple task automation to complex decision support and predictive analytics. • Drastically reduces overhead in billing and patient record management, allowing clinicians to focus on direct patient care. • High initial costs, data security concerns, and the need for specialized technical training remain the primary hurdles to adoption.
Paula R. Patel, Samuel Bechmann.	Discharge Planning	• Success depends on seamless collaboration between doctors, nurses, social workers, and families to prevent care gaps. • Discharge planning must begin at the time of admission to reduce hospital "length of stay" and improve bed turnover. • Identifying high-risk patients (elderly or those with chronic conditions) early is

		the most effective way to lower 30-day readmission rates.
Ribeiro Jorge, Lima Rui, Eckhardt Tiago, Paiv Sara	Robotic Process Automation and Artificial Intelligence in Industry 4.0 – A Literature review	• RPA acts as a non-invasive "bridge," connecting old industrial legacy systems with modern digital Industry 4.0 environments. • Integrating AI allows RPA to handle unstructured data (like images or text), expanding its use beyond simple repetitive tasks. • Automating back-office and shop-floor admin tasks increases organizational flexibility and speeds up the supply chain.
Wewerka Judith, Reichert Manfred	Towards Quantifying the Effects of Robotic Process Automation	• Most organizations lack a consistent framework or KPIs to accurately measure the actual ROI and long-term effects of RPA. • While time-saving is the primary goal, the most significant quantifiable benefit is often the total elimination of human transcription errors. • Effectiveness should be measured across three specific dimensions: Time (speed), Cost (resources), and Quality (accuracy).
Huang Wei-Lun , Liao Shu-Lang , Huang Hsueh-Ling , Su You-Xuan, Jerng Jih-Shuin , Lu Chien-Yu , Ho Wei-Sho, Xu Jing-Ran	A case study of lean digital transformation through robotic process automation in healthcare	• The study successfully combines Lean Six Sigma (DMAIC) with RPA to first identify "waste" in medical claim processes before automating them. • Post-implementation, the medical expense claim process time was reduced by 380 minutes, with Process Cycle Efficiency (PCE) jumping from 69% to 95%. • By automating repetitive manual data entry, the hospital was able to reallocate staff to high-value, patient-centric tasks that require human creativity and empathy.

Table 6: Literature Review Summary

Appendix B – Hospital Data Collection Support Letter

ESOFT METRO CAMPUS (PVT) LTD

Shaping Lives, Creating Futures

Head Office : No. 03, De Fonseka Place, Colombo 4, Sri Lanka.
Hotline : +94 11 7 572 572 | Fax : +94 11 7 575 462 | Email : info@esoft.lk | Web : www.esoft.lk

12th June 2025,

To whom it may concern,

Request for support to Conduct Research and Collect Data

I am writing to request your support on behalf of my student, Ms. Pawandi Wijegunawardane (K2377337), who is currently enrolled in the MSc in IT & Strategic Innovation program at Kingston University, affiliated with ESOFT Metro Campus, Colombo.

Project Name: **"Implementing Robotic Process Automation in Sri Lankan Private Healthcare to Enhance Efficiency and Effectiveness in the Patient Discharge Process."**

As part of her research, Ms. Pawandi Wijegunawardane intends to gather data through questionnaires with professionals and representatives from your esteemed organization. The insights and data collected will be invaluable in analyse the current discharge process, identify bottlenecks, and propose an RPA-based solution tailored to the Sri Lankan healthcare context will lead to improvement of the above-mentioned research topic.

We kindly request your support in allowing Ms. Pawandi Wijegunawardane to gather data for her research under the regulations of your institutional guidelines and observation. The findings from this study will contribute to the advancement of knowledge in the subject area.

Your assistance in facilitating this research will be greatly appreciated. Should you require further details or clarification, please do not hesitate to contact me.

Please note that this letter is issued upon the student's request (Ms. Pawandi Wijegunawardane). ESOFT Metro Campus will not bear any responsibility for the data collected during the course of the research.

Thank you for your consideration.

Yours sincerely,

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Mr. Kavindu Yakupitiya
Deputy Center Manager Academic Affairs
ESOFT Metro Campus, No 03, De Fonseka Road, Colombo 4
Email: kavindu.yakupitiya@esoft.lk
Contact Number: 0762984479

Appendix C – Staff Questionnaire

Project Title

Implementing Robotic Process Automation in Sri Lankan Private Healthcare to Enhance Efficiency and Effectiveness in the Patient Discharge Process

Researcher

This study is part of a master's level dissertation for the MSc in IT and Strategic Innovation program at **Kingston University London**.

Purpose of the Study

The objective of this academic research is to explore how the implementation of **Robotic Process Automation (RPA)** can improve the efficiency, speed, and accuracy of the **patient discharge process** in Sri Lanka's private hospitals. Discharge is a critical part of hospital operations, and this study aims to identify the current workflow, existing bottlenecks, and potential areas where automation can add value.

Why You Are Being Asked to Participate

You are being invited to complete this questionnaire because of your direct or indirect involvement in the patient discharge process. Your experience and insights will be instrumental in understanding the current operational practices and the feasibility of automation.

What Will Happen

You will be asked a series of multiple-choice and short-answer questions related to the discharge process in your hospital. The questionnaire should take around **10–15 minutes** to complete.

Confidentiality and Anonymity

- **No hospital names, staff names, or any personally identifiable information will be collected, used, or disclosed** in the final dissertation.
- All responses are **strictly confidential** and will be used **solely for academic research** purposes.
- Data will be stored securely and anonymously, and your participation will not affect your employment or professional standing in any way.

Voluntary Participation

Your participation is entirely voluntary. You may skip any question or stop at any time. Completion of the questionnaire will be considered as providing consent to use the responses for academic purposes.

Benefit of the Study

The findings of this study may be used to develop **practical recommendations and best**

practices for hospitals to improve discharge processes using modern automation technologies like RPA.

Contact for Further Information:

If you have any questions about this study or your participation, you may contact the researcher directly: **k2377337@kingston.ac.uk**

A university confirmation letter is also provided as a separate document for your reference.

Section 1: General Information

1. Which department do you work in?

☐ Nursing
☐ Medical (Physician)
☐ Pharmacy
☐ Billing / Finance
☐ Front Office / Administration
☐ Medical Records / IT
☐ Other (please specify): ___________________________

2. How many years have you worked in a hospital setting?

☐ Less than 1 year
☐ 1–3 years
☐ 4–7 years
☐ Over 7 years

3. How often are you involved in patient discharge processes?

☐ Daily
☐ Few times a week
☐ Occasionally
☐ Never (skip to end)

4. On average, how much time do you personally spend on tasks related to the patient discharge process per discharge?

☐ Less than 30 minutes
☐ 30 minutes – 1 hour
☐ 1 hour – 2 hours
☐ More than 2 hours

Section 2: Discharge Process Workflow

5. Please outline the typical steps you perform in the discharge process (you can list or explain):

6. Which of the following tasks do you handle during patient discharge? *(Tick all that apply)*

☐ Preparing medical discharge summaries
☐ Processing insurance documentation
☐ Communicating with pharmacy
☐ Informing caregivers or patients
☐ Coordinating billing and payments
☐ Updating discharge in the hospital system
☐ Notifying other departments (e.g., housekeeping, transport)
☐ Other: ____________________________

7. In your opinion, which step(s) cause the most delay in the discharge process?

☐ Final medical approval
☐ Documentation errors
☐ Inter-department coordination
☐ Insurance clearance
☐ Billing finalization
☐ Pharmacy coordination
☐ File clearance from departments
☐ Other: ___________________________

8. On average, how long does a complete discharge take once the doctor has approved it?

☐ Less than 1 hour
☐ 1–2 hours
☐ 2–4 hours
☐ More than 4 hours

9. Do you use any digital systems during the discharge process (e.g., HIS, billing, pharmacy systems)?

☐ Yes
☐ No

If yes, please mention the name of the system/software used:

Section 3: RPA Awareness and Automation Potential

Robotic Process Automation (RPA) is a technology that uses software robots to do simple, repetitive manual computer tasks just like a human would. In healthcare, this could mean entering patient information, sending appointment reminders, or handling billing.

10. Have you heard of Robotic Process Automation (RPA) before this study?

☐ Yes
☐ No

11. Do you believe parts of the discharge process can be automated?

☐ Yes
☐ No
☐ Not sure

12. Which of these tasks do you think can be effectively automated? *(Tick all that apply)*

☐ Discharge summary generation
☐ Insurance form processing
☐ Final bill calculation and approval
☐ Updating HIS system fields
☐ Pharmacy notifications
☐ Patient notification (SMS/email)
☐ Appointment scheduling for follow-up
☐ Other: ____________________________

13. What challenges do you see in implementing automation/RPA in the hospital's discharge process?

14. Would you be supportive of testing an automated solution (like RPA) in your department?

☐ Yes
☐ No
☐ Maybe (if well explained and trained)

Section 4: Final Comments

Please share any suggestions or feedback you have about improving the discharge process at your hospital:

Appendix D – Patient Survey

Patient Discharge Experience in Sri Lankan Private Hospitals – Feedback Form
I am a postgraduate student currently reading for the MSc in Information Technology and Strategic Innovation at Kingston University London.
As part of my final dissertation, I am conducting a research study titled:
"Implementing Robotic Process Automation in Sri Lankan Private Healthcare to Enhance Efficiency and Effectiveness in the Patient Discharge Process."
This study aims to understand the current experiences and challenges faced by patients during the discharge process in private hospitals in Sri Lanka. Your feedback will help identify common delays, pain points, and perceptions, which will be used to propose a technology-based solution to improve hospital discharge efficiency.
The questionnaire is intended for individuals who have been recently discharged or have assisted a family member or loved one during a discharge from a Sri Lankan private hospital.
Please note:

- Your participation is completely voluntary.
- All responses will remain anonymous and confidential.
- No personally identifiable information will be collected.
- The data will be used solely for academic research purposes.

Your honest input will be greatly appreciated and will play a key role in making valuable contributions to the improvement of healthcare processes in Sri Lanka.
Thank you for your time and support!

Section 1: Background Information
1.When was the most recent private hospital discharge you experienced (either yourself or a family member)?
Within the last month
    1–6 months ago
    6–12 months ago
    Over 1 year ago
2.Was the discharge for you or a family member?
    Myself
    Family member
3.Where in Sri Lanka did the discharge happen? *(City or District name)*
4.Age of the patient at discharge (approximate)
    Below 18
    18–30
    31–50
    51–70
    Over 70
5.What type of treatment or admission was the reason for your hospitalization? *(Select the most relevant option)*
    Surgical procedure

Medical treatment (e.g., fever, infections, chronic conditions)
Maternity
Diagnostic / Observation only
Accident or Emergency admission
Prefer not to say

Section 2: Discharge Experience

6.Approximately how long did you wait to be discharged after the doctor said you could go home?

Less than 1 hour
1–2 hours
3–5 hours
More than 5 hours

7.What took the most time during the discharge process? *(Select all that apply.)*

Waiting for bill
Waiting for pharmacy
Getting discharge documents
Approvals between departments
Insurance Clearance

8.Did the staff clearly explain medications and next steps before leaving?

Yes
Somewhat
No

9.Did the discharge process feel smooth and organized?

Yes
Somewhat
No

10.Do you think that digitalizing the discharge process will improve quality and efficiency of the discharge process?

Yes
No
Maybe

Section 3: Final Thoughts

11.Would you support a hospital that will use more automation or digital tools to speed up the discharge process?

Yes
Maybe
No

12.Any comments or suggestions to improve the discharge experience? *(Optional – Short Answer)*